Chiral constructions and regular subgroups in dimension three

Principal synthesis for Leemans–Vandenschrick

We recover the three principal theorems of *On chiral polytopes having a group* $\mathrm{PSL}(3,q)$ *as automorphism group*, in its fixed 25-page author edition. The constructions and their original matrices are due to Leemans and Vandenschrick. The accompanying papers give complete new proofs, distinguish their classical inputs, and record several corrections to intermediate formulas and tables. The final unitary examples in Section 7 are supplied with explicit generators. Earlier computational statements cited there remain attributed prior results, with their evidence boundary stated below.

The common ideas are the actual translation kernels of small projective subgroups, elementary matrix generation, and the action of an involution's centre and axis. A subgroup order does not by itself prove an intersection property: the constructions use the actual common subgroups, and the finite unitary examples check every indexed intersection. A separate centralizer calculation excludes reversing automorphisms of the whole ambient group.

The three principal conclusions

Let $q = p^f$, and let $d = \gcd(3, q-1)$. We use the standard string rotation correspondence for abstract regular and chiral polytopes.

Theorem 1. *The following assertions hold.*

1. *For every $q \geq 5$, the three matrices in Theorem 1.1 of the source, with a primitive element $x \in \mathbb{F}_q^\times$, generate a self-dual chiral polytope with automorphism group $\mathrm{PSL}_3(q)$ and type*

$$[q-1,\ 2(q-1)/d,\ q-1].$$

   The adjacent subgroups are the full set stabilizers of two distinct lines and two distinct points, respectively, of order $2q^2(q-1)^2/d$.

2. *For every prime $p \equiv 1 \pmod 6$, each of the four markings specified in Theorem 1.2 of the source gives a chiral polytope with automorphism group $\mathrm{PSL}_3(p)$. Their types are $[3,6,6,3]$ for the positive sign and $[3,6,3,3]$ for the negative sign. Including duals there are six distinct oriented marking classes in this construction, which give three unbased polytope classes after identifying the two opposite base-flag orbits. The same matrices over $\mathbb{F}_{p^2}$, with $p \equiv 5 \pmod 6$, give directly regular examples: $\mathrm{PSL}_3(p^2)$ is the rotation group and its extension by Frobenius is the full group.*

3. *For every $q \geq 2$, neither $\mathrm{PSL}_3(q)$ nor $\mathrm{PSU}_3(q)$ is the automorphism group of a chiral polytope of rank greater than five.*

Proof. The first accompanying paper, *Two stabilizers and elementary matrices in dimension three*, retains the literal source matrices and duality. Its finite-field span argument generates both full translation radicals, hence all six elementary root subgroups. This proves ambient generation without a maximal-subgroup classification. The stabilizer intersection gives the rotation intersection property. Its final semilinear calculation includes the inverse-transpose coset and proves chirality, while direct multiplication gives the type and self-duality. This proves the

first assertion for every prime power in the stated range, rather than only the fields used in supplementary controls.

A parabolic subgroup and a conic in the rank-five construction proves the exact translation module of the first adjacent subgroup and the restricted parabolic containing it. The two signs are treated separately. For the negative sign a literal symmetric-square representation gives the conic end; the positive sign uses the stated point-pair duality. Both actual middle intersections and every recursive rotation intersection are proved. The common centralizer of three prefix involutions is scalar, excluding all reversing automorphisms over the prime field. Frobenius reverses the quadratic-field markings. The comparison of both cube roots and both signs, including duals, gives exactly the oriented-class statement above. This proves the second assertion with its orientation and full regular-group qualifications.

For the third assertion, the paper *The two regular ends of a chiral marking* first derives the two directly regular ends from the actual indexed rotation intersection property. The acquired regular subgroup classification bounds their ranks. A rotation of order two gives a proper normal direct product, so it cannot occur in the relevant simple ambient groups. The elementary unitary argument for $\mathrm{PSU}_3(2)$ pays the nonsimple boundary separately. It remains to exclude rank six. In even characteristic the two modules and their actual centres and axes force a pair of distant rotations not to commute. In odd characteristic the Heisenberg normal form gives the same contradiction by an explicit commutator. The remaining directly regular simplex case has rotation group a quotient of A_7; group orders exclude it in all ambient fields of characteristic five. The complete argument is uniform and proves the claimed bound, including all its small-field premises. □

The auxiliary classifications and source tables

The source's Section 5 is recovered by four accompanying components. *Small regular subgroups and their translation kernels* proves the rank-two order classification and the complete odd-characteristic alternatives in rank three. *Elations and two translation modules* proves the even-characteristic normal forms, the exact field and intersection conditions, and the regular rank bound. *Four homologies in a projective plane* gives the odd rank-four classification, including its degenerate alternative. The conic branch reuses the already proved projective-line classification: the exceptional markings are those for $\mathrm{PSL}_2(11)$, $\mathrm{PSL}_2(19)$ and $\mathrm{PGL}_2(5)$. Their exact presentations and checks remain acquired inputs.

These proofs use Dickson's subgroup types and the classical three-dimensional conic representation as attributed inputs. They do not count those classifications as new results. In the odd affine branch, writing $K = \mathbb{F}_p(a)$ identifies the actual kernel as K^2; the complements are literal translates, so their intersections can be calculated before appealing to the string end criterion. In even characteristic the corresponding two modules determine precisely when the marking has the intersection property. This accounts for all group alternatives and types used in Tables 2–3. The earlier rank-three projective-line classifications cited by those tables remain attributed inputs, with the actual string and intersection hypotheses retained. No automatic passage from an arbitrary hypermap marking is used.

The separate calculation *The parameter restrictions in Table 1* pays all nine signed parameter rows of that auxiliary table. It computes the three characteristic polynomials and eliminates t from $t^2 = 5$ and the exact middle-order equations. The listed primes are necessary parameter restrictions; the actual square root must belong to the ambient field and satisfy the displayed equation. Neither the table nor its heading is used as an independent intersection certificate.

The source corrections have precise limits. The first elementary-matrix word on physical page 6 needs an x^{-1} factor in one coordinate; the generation paragraph on page 7 interchanges its end-subgroup names. In Section 4 the conic sign requires its own intersection argument; the adjacent translation kernel over a quadratic field has order p^2, and the regular analogue has an

additional Frobenius automorphism. The six-class sentence is qualified by the oriented-marking convention.

On page 11 a regular unipotent has order p in odd characteristic, not the printed bound $2(2,p)$. The final dihedral divisibility bound survives. On page 17 the direct product is D_{2p}^2, not D_{2p}; the final characteristic-polynomial coefficient is a', not b. The point/line statements are qualified by duality. The simplex alternative needed in Section 6 occurs in characteristic five, allowing embeddings into extension fields, rather than only the single ambient field of order five.

Finally, the generic affine row of Table 3 needs a type correction. If $\lambda+\lambda^{-1}=2a$, put $h=|\lambda|$ and $h'=|-\lambda|$. Its actual types are

$$[h,h'], \quad [h,2p], \quad [h',2p],$$

not just the printed $[k,4]$ and $[k,k]$. For example $p=5, a=2$ gives three C-markings in the group of order 300, with types $[3,6]$, $[3,10]$ and $[6,10]$. The uniform proof and full finite checks agree. This repairs a row without changing the principal constructions or the rank obstruction.

Section 7: positive examples and retained prior results

Three explicit unitary examples in rank four recovers the three positive $\mathrm{PSU}_3(17)$ examples reported in the source, with types $[3,7,3]$, $[3,8,3]$ and $[3,9,3]$. Its matrices are defined by three Hermitian centres over $\mathbb{F}_{17}[\beta]$, $\beta^2=3$. Fresh exact permutation calculations prove full ambient generation and all 768 indexed intersections. A written calculation of the full semilinear centralizer excludes every reversing automorphism. This is a positive existence statement; no absence of other types is inferred from the source's sentence or from the exploratory search.

The negative lists for $\mathrm{PSL}_3(q)$ with $q\le 4$ and for $\mathrm{PSU}_3(q)$ with $q\in\{2,3,4,5,7,9,13,16\}$ are explicitly attributed in the source to earlier computational work. The three $\mathrm{PSU}_3(11)$ pairs, of types $[5,4,5]$ twice and $[4,5,4]$ once, are likewise attributed there to data collected for Buekenhout, Leemans and Tranchida. We retain these as prior source-attributed results, not as our independently repeated censuses. The available edition of that earlier paper provides timings and group data but not those output tuples; the project's accepted algorithmic recovery does not manufacture the missing execution receipts. This evidence boundary is explicit in the retained-input register.

The same paragraph's negative list for $\mathrm{PSL}_2(q)$ contains one incorrect entry: $q=81$. The acquired projective-line theorem gives a chiral example of type $[40,40,40]$. Indeed its pointed construction applies with $p=3$, $d=4$ and $k=40$: k is even and divides $(81-1)/2$, while none of $3^e-1, 3^e+1$ for $1\le e<4$ is divisible by 40. This corrects the quoted list by an already proved theorem. The remaining entries of that list satisfy its negative criteria.

The previously known absence of chiral polyhedra for $\mathrm{PSL}_3(q)$ and $\mathrm{PSU}_3(q)$ is the theorem attributed to Breda d'Azevedo and Catalano. Together with Theorem 1(3), it gives the source's conclusion that a unitary chiral example can have only rank four or five. The other introductory classification statements are retained with their earlier authorship; the already acquired project results supply exact reuse links where available.

The historical question about unitary groups for $q>17$ is recorded without a new claim about its present status. The quoted composite-degree PSL_2 question is settled by the previously acquired coprime-subfield construction in the projective-line paper. Its full proof and earlier attribution are retained.

Two stabilizers and elementary matrices in dimension three

Human Compression project

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We reprove the rank-four construction of Leemans and Vandenschrick [1, Theorem 1.1]. Its matrices and duality are retained exactly. The generation argument below uses elementary matrices; it does not require a classification of maximal subgroups. We retain the classical rotation-group correspondence and its rank-four intersection criterion [2], as already stated explicitly in the project's common theory for projective markings. We also retain the classical description of automorphisms of $\mathrm{PSL}_3(q)$ as projective semilinear maps and their compositions with inverse transpose, for $q \geq 5$.

Let $F = \mathbb{F}_q$, where $q \geq 5$, and let x generate $F^\times$. Put

$$a = \begin{pmatrix} x & 1 & 0 \\ 0 & 1+x^{-1} & -x^{-1} \\ 0 & 1 & 0 \end{pmatrix}, \quad b = \begin{pmatrix} -x^{-1} & 0 & 0 \\ 0 & 0 & 1 \\ 0 & x & 0 \end{pmatrix}, \quad c = \begin{pmatrix} x & 0 & 0 \\ x-x^{-1} & 1 & 0 \\ x-x^{-1} & 0 & x^{-1} \end{pmatrix}.$$

All have determinant one. We use the same letters for their projective images. Set $d = \gcd(3, q-1)$.

Theorem 1. *The ordered triple (a, b, c) generates $\mathrm{PSL}_3(q)$ and defines a self-dual chiral polytope of type*

$$[q-1,\ 2(q-1)/d,\ q-1].$$

Its two end subgroups are the full set stabilizers of two distinct lines and two distinct points, respectively. Each has order $2q^2(q-1)^2/d$.

1 The full stabilizers

For $u, v \in F$ and $r \in F^\times$, write

$$K(u,v) = I + uE_{21} + vE_{31}, \qquad h(r) = \mathrm{diag}(r^{-2}, r, r).$$

Thus $K(u,v)K(u',v') = K(u+u', v+v')$ and $b^{2m} = h(x^m)$. Direct multiplication gives, for $r, s \in F^\times$,

$$h(r)c^{-1}h(s/r)ch(s^{-1}) = K(C/x, C), \quad C = (r^3 - s^3)(1-x^2),\ c^{-1}K(u,v)c \ = K(xu, x^2v). \quad (1)$$

The factor x^{-1} in the first coordinate of the first identity is essential. The corresponding displayed word on source page 6 omits it.

The $\mathbb{F}_p$-span V of all $r^3 - s^3$ is stable under multiplication by x^3 and contains $x^3 - 1 \neq 0$. Moreover $\mathbb{F}_p(x^3) = F$: otherwise x^3 would belong to a proper subfield of size at most $\sqrt{q}$, whereas

$$|x^3| = (q-1)/\gcd(3, q-1) \ \geq \ (q-1)/3 \ > \ \sqrt{q} - 1.$$

The strict inequality holds for $q > 4$. Hence V is a nonzero vector space over F, so $V = F$. Since $1 - x^2 \neq 0$, the first identity and addition in K give every $K(t/x, t)$. Conjugation by c

gives every $K(t, x^2 t)$. The vectors $(x^{-1}, 1)$ and $(1, x^2)$ are independent, with determinant $x - 1$. It follows that

$$K(F, F) \;\leq\; \langle b, c \rangle.$$

Let $p = [0, 1, 0]^t$, $p' = [0, 0, 1]^t$. Their full set stabilizer in $\mathrm{SL}_3(F)$ consists of the matrices

$$\begin{pmatrix} \alpha & 0 & 0 \\ u & \beta & 0 \\ v & 0 & \gamma \end{pmatrix}, \quad \alpha\beta\gamma = 1, \qquad \text{and their products with } b.$$

The subgroup fixing both points is $K(F, F)$ times the determinant-one diagonal torus. Modulo K, the diagonals of $b^{2m} c^n$ have exponent vectors

$$(-2m + n,\; m,\; m - n).$$

Every triple of exponents summing to zero occurs: for exponents (A, B, C) take $m = B$ and $n = A + 2B$. Thus b, c generate the full set stabilizer, even before taking projective images. Dividing its order by the scalar center gives $2q^2(q-1)^2/d$.

Put

$$D = \begin{pmatrix} (x - x^{-1})^{-1} & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & x \end{pmatrix}, \qquad \phi(g) = D(g^{-1})^t D^{-1}.$$

The matrix D is symmetric and invertible. Consequently $\phi^2 = 1$, and direct multiplication gives

$$\phi(a) = c^{-1}, \qquad \phi(b) = b^{-1}, \qquad \phi(c) = a^{-1}.$$

It interchanges $\langle b, c \rangle$ and $\langle a, b \rangle$. The latter is therefore the full stabilizer of the unordered pair of lines

$$\ell : y - z = 0, \qquad \ell' : y - x^{-1} z = 0.$$

One can also identify these lines directly from the actions of a, b; a fixes them and b exchanges them.

2 Generation and intersection

Let $\widetilde{G} = \langle a, b, c \rangle \leq \mathrm{SL}_3(F)$. The preceding argument gives all $I + tE_{21}$ and $I + tE_{31}$ in $\widetilde{G}$. Since ϕ preserves $\widetilde{G}$, it also gives the images of all these matrices under ϕ. The block diagonal shape of D shows that these images range over all

$$I + uE_{12} + vE_{13}.$$

Indeed inverse transpose changes $K(u, v)$ into $I - uE_{12} - vE_{13}$, and conjugation by D applies an invertible linear map to (u, v). Commutators now give the remaining root groups:

$$[I + sE_{21}, I + tE_{13}] = I + stE_{23}, \qquad [I + sE_{31}, I + tE_{12}] = I + stE_{32},$$

where $[g, h] = ghg^{-1}h^{-1}$. These six elementary root groups generate $\mathrm{SL}_3(F)$ by row elimination. This proves generation without any overgroup classification.

Write $H = \langle b, c \rangle$ and $A = \langle a, b \rangle$ in the projective group. An element of $A \cap H$ preserves both unordered pairs $\{p, p'\}$ and $\{\ell, \ell'\}$. It fixes $e = [1, 0, 0]^t = \ell \cap \ell'$, and hence has a diagonal or a monomial representative

$$\mathrm{diag}(\alpha, \beta, \gamma), \qquad \begin{pmatrix} \alpha & 0 & 0 \\ 0 & 0 & \beta \\ 0 & \gamma & 0 \end{pmatrix}.$$

On the line pp', the two distinguished intersections with ℓ, ℓ' have ratios $z/y = 1, x$. A diagonal representative preserves this pair only when $\gamma/\beta = 1$, since $x^2 \neq 1$. The determinant then makes it $h(\beta)$, a power of b^2. A monomial representative sends z/y to $\gamma/(\beta(z/y))$; pair preservation forces $\gamma/\beta = x$. Its determinant makes it $h(\beta)b$, again a power of b. Thus $A \cap H = \langle b \rangle$.

Every power of c fixes p, p' separately. An element shared with $\langle b \rangle$ must therefore be an even power of b, acting as a scalar on pp'. But c^n acts there with eigenvalue ratio x^n, so it is shared only if $q - 1$ divides n. The eigenvalues of c are the distinct elements $x, 1, x^{-1}$; hence its projective order is $q - 1$ and this shared element is the identity. Applying ϕ gives the other adjacent cyclic intersection. We have proved

$$\langle a \rangle \cap \langle b \rangle = 1, \quad \langle b \rangle \cap \langle c \rangle = 1, \quad A \cap H = \langle b \rangle.$$

Direct multiplication gives $(ab)^2 = (bc)^2 = (abc)^2 = I$; each product is nonscalar. The retained rank-four criterion now proves the full rotation intersection property.

3 Excluding reversal

Suppose an automorphism α of $\mathrm{PSL}_3(F)$ sends a to a^{-1} and fixes ab and abc. Then it fixes $c = (ab)^{-1}(abc)$ and preserves $A = \langle a, ab \rangle$.

The group H fixes no projective point. Indeed its subgroup $K(F, F)$ fixes precisely the points on pp'; the full diagonal torus has only p, p' as common fixed points there, and b exchanges them. Dually, A fixes no line, although it fixes e. A graph automorphism followed by a semilinear collineation would take A to a group fixing a line. It therefore cannot preserve A. By the retained automorphism theorem, α is induced by a semilinear collineation with field part $t \mapsto t^{p^i}$, $0 \leq i < [F : \mathbb{F}_p]$.

The axes of ab and abc are respectively

$$z = 0, \qquad (1 - x)y_1 = z,$$

where y_1 denotes the first homogeneous coordinate. These follow, in both characteristics, from their displayed matrices; their axes are uniquely determined by the involutions. Their intersection is p, so the collineation fixes p. It also permutes the three eigenpoints of a, namely

$$[1, 0, 0]^t, \quad [1, 1 - x, 1 - x]^t, \quad [1, x^{-1} - x, 1 - x^2]^t.$$

Only the first lies on the invariant line $z = 0$, so it fixes e. The linear part P consequently has the form

$$P = \begin{pmatrix} u & 0 & w \\ 0 & v & t \\ 0 & 0 & s \end{pmatrix}.$$

Projective centralization of c means $Pc^{(p^i)}P^{-1} = \mu c$ for a scalar μ. Applying this equation to p, whose eigenvalue is one on both sides, gives $\mu = 1$. The other two eigenvalues then imply $x^{p^i} = x$ or x^{-1}. The second option is impossible: for $0 \leq i < [F : \mathbb{F}_p]$ and $q \geq 5$, $p^i + 1 < q - 1$. The first option implies $i = 0$, since x is primitive. The three eigenpoints of c must therefore be individually fixed. They are p, p' and $[x, 1 + x, x]^t$. Fixing p' gives $w = t = 0$; fixing the last point gives $u = v = s$, as $x(1 + x) \neq 0$. Thus P is scalar and α is the identity. This contradicts $\alpha(a) = a^{-1}$, because a has order $q - 1 > 2$. The marking is therefore chiral by the retained correspondence.

Finally, a, c both have order $q - 1$. Odd powers of b interchange p, p' and cannot be scalar; $b^{2m} = h(x^m)$ is scalar precisely when $x^{3m} = 1$. Hence $|b| = 2(q - 1)/d$. The automorphism ϕ sends the marking to its dual (c^{-1}, b^{-1}, a^{-1}), proving self-duality and the claimed type.

4 Attribution and scope

The matrices, stabilizer descriptions, duality and rank-four existence result are prior work of the source authors. We supply the finite-field span argument, the full elementary generation proof and the detailed projective scalar check in the reversal argument. The source's displayed unipotent word needs the factor x^{-1} noted above. Its next-page transitivity paragraph also interchanges the two named end subgroups; the elementary generation argument avoids that paragraph entirely. These corrections do not change Theorem 1.1.

This component treats Theorem 1.1 and its proof, not the source's rank-five construction, subgroup classifications or global rank obstruction. It gives no new answer to an open question. The automorphism theorem and the polytope correspondence remain explicit classical premises; finite examples are controls on the formulas, not the proof of the all-field statement.

References

[1] D. Leemans and A. Vandenschrick, *On chiral polytopes having a group* $\mathrm{PSL}(3, q)$ *as automorphism group*, fixed 25-page author edition, 2022, Theorem 1.1 and Section 3.

[2] E. Schulte and A. I. Weiss, *Chiral polytopes*, Applied Geometry and Discrete Mathematics, DIMACS 4 (1991), 493–516; rotation correspondence and rank-four intersection criterion.

[3] Human Compression project, *Common theory for projective markings and rotations*, Topic E, version 3, Section 2; retained rotation correspondence.

A parabolic subgroup and a conic in the rank-five construction

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We give the subgroup and intersection arguments omitted from Section 4 of Leemans–Vandenschrick [1]. The four source matrices are unchanged. The two signs require different geometry: the second end fixes a point for one sign and preserves a conic for the other. We retain the classical rotation correspondence, its recursive intersection criterion [1, Lemmas 2.2 and 2.3], and the description of the automorphisms of $\mathrm{PSL}_3(F)$ as semilinear collineations and their compositions with inverse transpose. No classification of subgroups of $\mathrm{PSL}_3(F)$ or $\mathrm{PGL}_2(F)$ enters the proof.

Throughout, $p \geq 5$ is prime, w has order three, and $F = \mathbb{F}_p(w)$. Thus $F = \mathbb{F}_p$ when $p \equiv 1 \pmod 3$, and $F = \mathbb{F}_{p^2}$ when $p \equiv -1 \pmod 3$. Put $q = |F|$ and $k \in \{1, -1\}$, and set

$$a = \mathrm{diag}(1, w, w^2), \quad b = \begin{pmatrix} -1 & 0 & -k \\ 0 & -w^2 & -kw^2 \\ 0 & 0 & w \end{pmatrix}, \quad c = \begin{pmatrix} 1 & k & 0 \\ 0 & k & 1 \\ 0 & -1 & 0 \end{pmatrix}, \quad d = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}.$$

These matrices have determinant one. When discussing projective groups we use the same letters for their images. Write $e_1 = [1, 0, 0]^t$ and $\mu_6 = \langle -w^2 \rangle \leq F^\times$.

Theorem 1. *The marking (a, b, c, d) generates $\mathrm{PSL}_3(F)$, satisfies the string rotation relations and the full rotation intersection property, and has type $[3, 6, 6, 3]$ when $k = 1$ and $[3, 6, 3, 3]$ when $k = -1$. If $F = \mathbb{F}_p$, it defines a chiral polytope. If $F = \mathbb{F}_{p^2}$ with $p \equiv -1 \pmod 3$, it defines a directly regular polytope with rotation group $\mathrm{PSL}_3(F)$; its full automorphism group is the extension by the Frobenius automorphism, of order twice that of the rotation group.*

1 One projection and its translation kernel

We first work in $\mathrm{SL}_3(F)$. Let $\widetilde{A} = \langle a, b, c \rangle$. Every element has block form

$$\begin{pmatrix} \epsilon & v \\ 0 & M \end{pmatrix}, \qquad \epsilon \in \{1, -1\}, \quad \det M = \epsilon.$$

Projection to M will be denoted by π. Put $t = k(2w + 1)$; it is nonzero because $(2w + 1)^2 = -3$. Direct multiplication gives

$$u = b^2 a^{-1} = I + tE_{13} - tE_{23}.$$

The lower block of u is an upper unipotent matrix with parameter $-t$. Conjugation by the lower block of a multiplies that parameter by w^2. Since $F = \mathbb{F}_p(w)$, additive combinations give every upper unipotent matrix in $\mathrm{SL}_2(F)$. The lower block of c is

$$\begin{pmatrix} k & 1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & -k \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

Consequently the image contains the matrix interchanging the two root groups, and then all of $\mathrm{SL}_2(F)$ by elementary elimination. The lower block of b has determinant -1, so

$$\pi(\widetilde{A}) = \{M \in \mathrm{GL}_2(F) : \det M = \pm 1\}.$$

The kernel is an additive subgroup V of the row space F^2, normal under the natural action of $\mathrm{SL}_2(F)$. It cannot be zero. If it were, the unique lift of the central matrix $-I_2$ would be central in $\widetilde{A}$ and have the form

$$z = \begin{pmatrix} 1 & v \\ 0 & -I_2 \end{pmatrix}.$$

Commutation with a forces $v = 0$, since $w, w^2 \neq 1$. But $\mathrm{diag}(1, -1, -1)$ does not commute with c: its nonzero entry k in position $(1, 2)$ changes sign. This is impossible in odd characteristic. Thus $V \neq 0$. The action of $\mathrm{SL}_2(F)$ is transitive on nonzero row vectors, so normality forces $V = F^2$. We have proved

$$\widetilde{A} = \left\{ \begin{pmatrix} \epsilon & v \\ 0 & M \end{pmatrix} : \epsilon = \pm 1,\ \det M = \epsilon,\ v \in F^2 \right\}. \tag{1}$$

Its intersection with the scalar center of $\mathrm{SL}_3(F)$ is trivial. The projective image A therefore has order $2q^3(q^2 - 1)$. Equivalently, A consists of the projective transformations fixing e_1 whose eigenvalue there, in an $\mathrm{SL}_3(F)$ representative, belongs to μ_6. Multiplication by a central cube root changes that eigenvalue to 1 or -1. In particular A is generally smaller than the full point stabilizer.

Equation (1) supplies all $I + sE_{12}$ and $I + sE_{13}$. Conjugation by d cycles the coordinate indices and supplies the other four elementary root groups. Hence

$$\langle a, b, c, d \rangle = \mathrm{SL}_3(F)$$

before taking projective images.

2 The first adjacent pair

Let $T = \{I + sE_{13} + tE_{23} : s, t \in F\}$ and let $D_0 = \mathrm{diag}(w, w^2)$ act on its two parameters. The vector of $u = b^2a^{-1}$ is $t(1, -1)$. Its orbit spans exactly two dimensions over $\mathbb{F}_p$: the vectors $t(1, -1)$ and $t(w, -w^2)$ are independent, and $D_0^2 + D_0 + I = 0$. Denote this additive subgroup by V_0. Conjugation by b acts on the parameters as $-D_0^2$, so preserves V_0; conversely its square acts as D_0. Since $a = u^{-1}b^2$,

$$H = \langle a, b \rangle = V_0 \rtimes \langle b \rangle, \qquad |H| = 6p^2. \tag{2}$$

Here b has order six, and its nonidentity powers have nontrivial diagonal image. When $F = \mathbb{F}_p$, $V_0 = T$; when $F = \mathbb{F}_{p^2}$, it has order q, not q^2. All elements of H fix both e_1 and e_2.

For $1 \leq j \leq 5$ the entry $(b^j)_{13}$ is nonzero. Indeed it is

$$-k\,\frac{(-1)^j - w^j}{-1 - w},$$

and $-w^2$ has order six. Thus no such power is diagonal, proving $\langle a \rangle \cap \langle b \rangle = 1$ even projectively.

3 The point case

Suppose $k = 1$ and put $r = [1, w, w^2]^t$. Both b, c fix e_1 and r. The source matrix

$$J(w) = \begin{pmatrix} w & w^2 & 1 \\ w^2 & w & 1 \\ 1 & 1 & 1 \end{pmatrix}$$

has determinant $-3(2w+1) \neq 0$. Literal multiplication gives

$$J^{-1}a(w)J = w^2d^{-1}, \qquad J^{-1}b(w)J = wc(w^2)^{-1},$$
$$J^{-1}c(w)J = w^2b(w^2)^{-1}, \qquad J^{-1}dJ = wa(w^2)^{-1}.$$

Thus it exchanges the marking with the inverse-reversed marking at w^2, projectively. Applying (1) at w^2 shows that $B = \langle b, c, d\rangle$ is the corresponding restricted stabilizer of r: its allowed eigenvalues at r are precisely μ_6.

Choose the basis (e_1, r, e_2), represented by

$$P = \begin{pmatrix} 1 & 1 & 0 \\ 0 & w & 1 \\ 0 & w^2 & 0 \end{pmatrix}.$$

In that basis the two middle generators are

$$b' = \operatorname{diag}(-1, w, -w^2), \qquad c' = \begin{pmatrix} 1 & 0 & -w^2 \\ 0 & -w^2 & -w \\ 0 & 0 & -w \end{pmatrix}.$$

Their commutator $b'c'b'^{-1}c'^{-1}$ has translation vector $(-(2w+1), w)$ in positions $(1,3),(2,3)$. Conjugation by b' acts on these two coordinates with multipliers w and $-w^2$.

The prime-field minimal polynomials of these multipliers are distinct and relatively prime. In the prime-field case this follows from their different values; in the quadratic case their roots have orders three and six, respectively. Polynomial projections in this action therefore separate the two coordinates. Each nonzero coordinate then generates all of F under its multiplier, since $F = \mathbb{F}_p(w)$. Consequently $E = \langle b, c\rangle$ contains the full two-dimensional translation group in this basis.

Modulo that group, the two eigenvalue pairs at (e_1, r) are $(-1, w)$ and $(1, -w^2)$. They generate all $\{\pm 1\} \times \mu_6$: the second pair has order six and cannot supply the first coordinate -1. They therefore give all twelve allowed diagonal classes. After choosing the unique scalar adjustment making the first eigenvalue ± 1, these are exactly the elements of $A \cap B$. We conclude that

$$A \cap B = E, \qquad |E| = 12q^2. \tag{3}$$

The action of V_0 on r, whose third coordinate is nonzero, is free. Thus the stabilizer of r in H has order at most six. It contains $\langle b\rangle$, so $H \cap E = \langle b\rangle$. Also $\langle b\rangle \cap \langle c\rangle = 1$: in the displayed basis b' is diagonal, whereas the $(1,3)$ entry of $(c')^j$ is nonzero for $1 \leq j \leq 5$, because its two diagonal values have ratio of order six. These statements, (2), and the rotation criterion prove the full intersection property for (a, b, c).

The same conclusion at w^2, transported by J, proves the intersection property for (b, c, d). In particular $E \cap \langle c, d\rangle = \langle c\rangle$. Combining this with (3) gives

$$A \cap \langle c, d\rangle = \langle c\rangle.$$

Finally $A \cap \langle d\rangle = 1$, since d cycles the three coordinate points and A fixes e_1.

4 The conic case

Suppose $k = -1$. The conic in the source is

$$xy + wyz + w^2zx = 0.$$

Set $U = x+wz$, $V = z$, $W = y+w^2z$. Its equation becomes $UW - V^2 = 0$, with parametrization $[s : t] \mapsto [s^2 : st : t^2]$. For $g = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}$, define

$$\iota(g) = \frac{1}{\det g} \begin{pmatrix} \alpha^2 & 2\alpha\beta & \beta^2 \\ \alpha\gamma & \alpha\delta + \beta\gamma & \beta\delta \\ \gamma^2 & 2\gamma\delta & \delta^2 \end{pmatrix}.$$

This is the symmetric-square representation divided by its determinant; it is a homomorphism into $\mathrm{SL}_3(F)$ and is unchanged by scalar multiplication of g. Its projective action is faithful on $\mathrm{PGL}_2(F)$: an element fixing every quadratic point fixes every point of the projective line and is scalar.

Let T denote the above coordinate change. Exact multiplication gives

$$w^2 TbT^{-1} = \iota \begin{pmatrix} -w^2 & 0 \\ 0 & 1 \end{pmatrix}, \quad wTcT^{-1} = \iota \begin{pmatrix} w & -w^2 \\ 0 & 1 \end{pmatrix}, \quad w^2 TdT^{-1} = \iota \begin{pmatrix} -w^2 & 1 \\ -w & 0 \end{pmatrix}.$$

The scalar factors have order three and disappear projectively. We may therefore work with these three 2×2 matrices, denoted by $\bar{b}, \bar{c}, \bar{d}$.

The first two act on an affine coordinate as

$$z \mapsto -w^2 z, \qquad z \mapsto wz - w^2.$$

Their commutator is a nonzero translation. Conjugation by their multipliers gives its entire F-span over $\mathbb{F}_p$, hence all translations, while the multipliers generate μ_6. Thus

$$\langle \bar{b}, \bar{c} \rangle = \{ z \mapsto tz + h : t \in \mu_6,\ h \in F \}, \qquad |E| = 6q. \tag{4}$$

The element $\bar{d}$ sends infinity to w. Conjugating the full translation group and then translating w to zero gives the full opposite unipotent root group. The two root groups generate $\mathrm{PSL}_2(F)$. The determinants of $\bar{c}, \bar{d}$ are w, a square; the determinant of $\bar{b}$ is $-w^2$. It follows directly from the determinant-square quotient that

$$B \cong \begin{cases} \mathrm{PSL}_2(F), & -1 \text{ is a square in } F, \\ \mathrm{PGL}_2(F), & -1 \text{ is not a square in } F. \end{cases}$$

No subgroup classification is needed for this conclusion.

Under the coordinate change, e_1 is the conic point infinity and e_2 is the point zero. An element of B fixing e_1 is represented by $\begin{pmatrix} t & h \\ 0 & 1 \end{pmatrix}$. Its eigenvalue at e_1 in the displayed determinant-one lift is t. By (1), it belongs to A exactly when $t \in \mu_6$. These multipliers all occur in (4), so once again

$$A \cap B = E.$$

Since H also fixes e_2, its intersection with E fixes zero in the affine model. This subgroup is exactly $\langle \bar{b} \rangle$; therefore $H \cap E = \langle b \rangle$. The affine transformation $\bar{c}$ has order three, and neither of its nonidentity powers fixes zero. Hence $\langle b \rangle \cap \langle c \rangle = 1$. Together with the first cyclic intersection, this proves IP^+ for (a, b, c).

The matrices c, d permute the four points

$$e_1, e_2, e_3, [1, 1, 1]^t$$

as two three-cycles: c fixes e_1, and d cycles e_1, e_2, e_3. These points form a projective frame, so the action is faithful and $\langle c, d \rangle \cong A_4$. Its stabilizer of e_1 is $\langle c \rangle$, giving

$$A \cap \langle c, d \rangle = \langle c \rangle, \qquad A \cap \langle d \rangle = 1.$$

This supplies the actual conic intersection argument, rather than the point-pair shorthand printed at the end of source Section 4.

5 The full intersection property and orientation

For either sign we have proved IP^+ for (a, b, c) and all three recursive intersections

$$A \cap \langle b, c, d\rangle = \langle b, c\rangle, \quad A \cap \langle c, d\rangle = \langle c\rangle, \quad A \cap \langle d\rangle = 1.$$

The retained recursive criterion therefore gives the full rank-five rotation intersection property. The six interval products are nonidentity projective involutions. Literal multiplication gives squares I for $ab, bc, cd, abc, abcd$, and $(bcd)^2 = w^2 I$. Their nontriviality is apparent from the nonzero off-diagonal entries. The eigenvalues of a, b, c, d give their projective orders $3, 6, 6, 3$ for $k = 1$ and $3, 6, 3, 3$ for $k = -1$.

Suppose first that $F = \mathbb{F}_p$. An automorphism reversing the marking must invert a and fix the three prefix involutions $ab, abc, abcd$. It preserves $A = \langle a, ab, abc\rangle$. This group fixes e_1 and fixes no line: its translation subgroup rules out lines not containing e_1, and its lower $\mathrm{SL}_2(F)$ rules out a fixed line through e_1. An automorphism involving inverse transpose would send its fixed point to a fixed line, so cannot preserve A. There is no nontrivial field automorphism. The proposed reverser is therefore conjugation by a matrix $P \in \mathrm{GL}_3(F)$.

Projective commutation with a determinant-one involution whose square is I is ordinary commutation: its scalar multiplier has both square and cube one, so equals one. Here

$$ab = \begin{pmatrix} -1 & 0 & -k \\ 0 & -1 & -k \\ 0 & 0 & 1 \end{pmatrix}, \quad abc = -P_{23}, \qquad abcd = -P_{12},$$

where P_{ij} is the coordinate transposition matrix. Commuting with P_{12}, P_{23} forces $P = \alpha I + \beta \mathcal{J}$, with $\mathcal{J}$ the all-ones matrix. The $(3, 1)$ entry of $\mathcal{J}(ab) - (ab)\mathcal{J}$ is -2, so $\beta = 0$. Thus P is scalar and cannot invert the element a of order three. This proves chirality over the prime field for both signs.

If $F = \mathbb{F}_{p^2}$ with $p \equiv -1 \pmod 3$, Frobenius sends w to w^2 and hence a to a^{-1}. All three displayed prefix involutions are defined over $\mathbb{F}_p$, so Frobenius fixes them. It is an involutory automorphism of the rotation group and is outer. The retained correspondence gives the directly regular extension by this automorphism; its full group, rather than its rotation subgroup, has order $2|\mathrm{PSL}_3(F)|$.

6 The six marking classes and attribution

Over $\mathbb{F}_p$, the two choices w, w^2 give different ordered marking classes for each sign. Indeed they are exactly the enantiomorphic markings: $a(w^2) = a(w)^{-1}$, $b(w^2) = a(w)^2 b(w)$, and c, d are unchanged. An isomorphism of these ordered markings would be the reverser just excluded. For $k = 1$ the matrix J shows that their two dual markings give the same two classes. For $k = -1$ the dual type is $[3, 3, 6, 3]$, different from both original types; it contributes two further classes. Thus the source construction gives six ordered, oriented marking classes, including duals.

The word "polytopes" in the source's count must be read with this orientation convention. The two enantiomorphic markings are the two choices of base-flag orbit in one unbased chiral polytope. If orientation is forgotten, the above six classes give three unbased isomorphism classes: one of type $[3, 6, 6, 3]$, and a dual pair of the other two types. The first is self-dual after orientation is forgotten. These distinctions follow from the retained rotation correspondence; they do not alter the source's existence theorem.

The matrices, the two signs, the conic, the duality matrix, and the prime and quadratic-field constructions are the source authors' work. The projection-kernel argument, the separated translation modules, the explicit symmetric-square model and the full recursive intersections above provide the omitted uniform proofs. We also make the full regular group and the orientation convention explicit. This component treats Theorem 1.2 and Section 4 only. It does not settle the source's later classifications, global rank obstruction, or open questions.

References

[1] D. Leemans and A. Vandenschrick, *On chiral polytopes having a group* $\mathrm{PSL}(3,q)$ *as automorphism group*, fixed 25-page author edition, 2022, Theorem 1.2 and Section 4.

[2] E. Schulte and A. I. Weiss, *Chiral polytopes*, DIMACS 4 (1991), 493–516; rotation correspondence and recursive intersection criterion, retained as in source Section 2.

Elations and two translation modules

A recovery of the even-characteristic classification in L027

Let F be a finite field of characteristic two. A string C-marking is an ordered independent generating tuple of involutions satisfying the string commutation relations and the subgroup intersection property. It is nondegenerate when every adjacent product has order greater than two. We prove the assertions of Leemans–Vandenschrick, Section 5.4–5.5, with the actual field and intersection conditions made explicit. The matrices in the normal forms are those of that source. The common geometric argument and the two-module proof below replace its separate calculations.

We retain the elementary projective interpretation of a linear action, the usual criterion which deduces the intersection property from the two end markings and their intersection, and Dickson's subgroup theorem in the following precise form. A subgroup of $\mathrm{PSL}_2(F)$ is cyclic, dihedral of order $2k$ with k odd, contained in a point stabilizer, or conjugate to a natural $\mathrm{PSL}_2(K)$ for a subfield $K \subseteq F$. These alternatives include the small groups; in particular A_5 is $\mathrm{PSL}_2(4)$, and an A_4 subgroup is in a point stabilizer. We also retain the standard faithful action of the stabilizer of a nonsingular plane conic as $\mathrm{PSL}_2(F)$. No subgroup multiplicity or source census is used.

Commutation and the flag case

Every projective involution has a unique order-two lift to $\mathrm{SL}_3(F)$, since the scalar centre has odd order dividing three. The lift has the form $I + uf$, where $u \neq 0$, $f \neq 0$ and $f(u) = 0$. Its centre is the point $\langle u \rangle$ and its axis is the line $\ker f$. If two projective involutions commute, their order-two lifts commute: an equality $AB = \zeta BA$ with scalar ζ implies $\zeta^2 = 1$.

Lemma 1. *Two distinct elations commute if and only if they have the same centre or the same axis.*

Proof. For $N = uf$ and $M = vg$, commutation is $uf(v)g = vg(u)f$. If both the vectors and the functionals are independent, this forces $f(v) = g(u) = 0$. The independent vectors u, v would then lie in the one-dimensional intersection of the distinct kernels, a contradiction. Conversely, a common centre or axis makes both products zero. □

Theorem 2. *A nondegenerate string C-marking whose group fixes a point and an incident line has rank at most three. The possibilities in ranks one, two and three are respectively C_2, D_8, and $C_2^2 \wr C_2$, of types* [], [4], *and* [4, 4].

Proof. Choose a basis along the fixed flag. Every generator is upper unitriangular. Write such a matrix as $(x, y, z) = I + xE_{12} + yE_{13} + zE_{23}$; it is an involution precisely when $xz = 0$. A generator with $x = z = 0$ is central and cannot occur in a nondegenerate marking of rank at least two. Noncommuting adjacent generators must alternate between $x \neq 0, z = 0$ and $x = 0, z \neq 0$. Their product has order four, giving D_8. At rank four the first and fourth generators are of opposite kinds and do not commute, contrary to the string relation.

At rank three, reverse the coordinates if necessary and write the ends as $r = (a, b, 0)$, $t = (e, f, 0)$, and the middle as $s = (0, c, d)$, with $a, e, d \neq 0$. Their commutators with s are

the central matrices $z_a = (0, ad, 0)$ and $z_e = (0, ed, 0)$. If $a = e$, the two dihedral ends share the nonidentity $z_a = z_e$, which is not in $\langle s \rangle$; the intersection property fails. If $a \neq e$, the four commuting involutions r, t, z_a, z_e are independent over $\mathbb{F}_2$: independence of a, e first kills the coefficients of r, t, and then that of ad, ed kills the remaining coefficients. They form a normal group of order 16. Conjugation by s sends r to rz_a and t to tz_e. It therefore interchanges the two bases (r, t) and (rz_a, tz_e), giving $C_2^2 \wr C_2$. The two dihedral end subgroups meet in $\langle s \rangle$, since their respective base spaces $\langle r, z_a \rangle$ and $\langle t, z_e \rangle$ are disjoint. This proves the intersection property and the asserted type. □

Rank three without an incident fixed flag

We write $M(x, y) = I + xE_{21} + yE_{23}$. The following small module calculation will be used twice. If $t \neq 0$, $K = \mathbb{F}_2(t)$, and the two involutions act on rows by

$$(x, y) \longmapsto (x, y + tx), \qquad (x, y) \longmapsto (x + y, y),$$

the module generated by $(0, c)$ is cK^2. Indeed the differences give $(c, 0)$ and $(0, tc)$; iteration gives every polynomial in t in both coordinates. The reverse inclusion follows from invariance. The product of the two matrices has trace t and determinant one. Its eigenvalues are λ, λ^{-1}, with $\lambda \neq 1$; its order $k > 1$ is odd, and $K = \mathbb{F}_2(\lambda + \lambda^{-1})$. Consequently k divides $|K| - 1$ or $|K| + 1$. The generated group is the actual dihedral group D_{2k}.

Theorem 3. *Let (r, s, t) be a rank-three string C-marking with no fixed incident point–line pair. Up to a projective change of coordinates and duality, its lifts have the form*

$$r = I + aE_{13}, \qquad s = I + E_{31}, \qquad t = I + bE_{13} + cE_{23}, \qquad a \neq 0.$$

These lifts fix e_2, so their group meets the scalar centre trivially. If $b = 0$ or $b = a$, then $c \neq 0$ and the group is $K^2 \rtimes D_{2k}$, where $K = \mathbb{F}_2(a)$ and the action is the natural two-dimensional dihedral action. The types are respectively $[k, 4]$ and $[k, k]$. Otherwise the group is $\mathrm{PSL}_2(K)$ for $K = \mathbb{F}_2(a, b)$.

Proof. The ends commute, so after duality they share an axis l. The axis m of s differs from l, since otherwise all three commute and fix a flag. Put $p = l \cap m$. The centre of s is not p: otherwise the whole group preserves (p, l). The centres of the two ends cannot both be p, since then (p, m) is fixed. Reversing the marking if necessary, take the centres of r, s and the point p as the three coordinate points. Scaling gives the displayed matrices.

If $b = 0$, the third generator is $M(0, c)$; if $b = a$, its product with r is $M(0, c)$. In either case $c \neq 0$ by nonidentity and independence. The group generated by r, s has trivial intersection with the translation group $M(F^2)$, and its action on that group is precisely the preceding module calculation. Thus the asserted semidirect product is an equality of actual matrix groups. When $b = 0$, the other dihedral end has order eight and intersects $\langle r, s \rangle$ in $\langle s \rangle$. When $b = a$, set $v = M(c/a, 0)$. Conjugation by v fixes s and takes r to t, so the other end is the conjugate of $D = \langle r, s \rangle$. Its intersection with D is the stabilizer in D of the nonzero vector $(c/a, 0)$. No nonidentity rotation fixes such a vector, and two distinct reflections fixing it would have a nonidentity rotation as their product. The stabilizer is therefore exactly $\langle s \rangle$. These arguments also prove the claimed types and the intersection property for these normal forms.

Now suppose $b \notin \{0, a\}$. If $c = 0$, the action on the plane $\langle e_1, e_3 \rangle$ is faithful. If $c \neq 0$, direct substitution shows that all three matrices preserve the nonsingular conic

$$Q = \frac{x_1^2 + ax_1x_3 + ax_3^2}{b(a + b)} + \frac{x_2^2}{c^2} = 0.$$

Its polar radical is $\langle e_2 \rangle$ and $Q(e_2) \neq 0$. The induced action on the quotient by this point is faithful: a kernel element has the form $M(x, y)$, and preservation of Q gives $(xx_1 + yx_3)^2/c^2 = 0$, hence $x = y = 0$. In either case the group is isomorphic to

$$H = \left\langle \begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix} \right\rangle \leq \mathrm{SL}_2(K), \qquad K = \mathbb{F}_2(a, b).$$

The first and third matrices are distinct commuting involutions, excluding the odd dihedral alternative. The first and second have different unique fixed points, excluding a point stabilizer. Dickson's theorem therefore gives a conjugate of $\mathrm{PSL}_2(K')$. The traces of the products of the first and second and of the second and third matrices are a, b. Conjugacy preserves trace, so $K \subseteq K'$; the group order bound inside $\mathrm{SL}_2(K)$ gives $|K'| \leq |K|$. Thus $K' = K$. The conclusion is a classification of groups of the stated C-markings; it does not assert that every remaining pair a, b has the intersection property. □

Rank four and the actual intersection

Theorem 4. *A nondegenerate rank-four string C-marking in* $\mathrm{PSL}_3(F)$ *is, up to coordinates and duality, of the form*

$$r_0 = I + E_{23}, \quad r_1 = I + aE_{31}, \quad r_2 = I + bE_{13}, \quad r_3 = I + E_{21}, \qquad ab \neq 0.$$

Put $K = \mathbb{F}_2(ab)$ *and let* k *be the order of* $r_1 r_2$*. These matrices satisfy the intersection property exactly when* $a \notin K$*, equivalently* $b \notin K$*. In that case their group is* $K^4 \rtimes D_{2k}$*, their two ends are* $K^2 \rtimes D_{2k}$*, and their type is* $[4, k, 4]$*. There is no nondegenerate marking of rank greater than four.*

Proof. The group cannot fix an incident flag by Theorem 2. After duality, r_0, r_2 share an axis l. If r_1, r_3 share a centre p on l, the whole group fixes (p, l); if their common centre is off l, then r_0, r_3 have neither a common axis nor a common centre and cannot commute. Thus r_1, r_3 share an axis $m \neq l$. Since r_0, r_3 commute, their common centre is $p = l \cap m$. Nondegeneracy forces the centres of r_1, r_2 to differ from p. Taking these three centres as coordinate points and scaling gives the displayed form.

Let $D = \langle r_1, r_2 \rangle$. On the coordinates (x_1, x_3) it is the dihedral group of order $2k$, with rotation trace ab; it meets $M(F^2)$ trivially. On this translation group conjugation is

$$(x, y)^{r_1} = (x + ay, y), \qquad (x, y)^{r_2} = (x, y + bx).$$

The module generated by $r_0 = M(0, 1)$ is $N_0 = M(aK, K)$, and that generated by $r_3 = M(1, 0)$ is $N_3 = M(K, bK)$. To see equality, successive differences first give the other coordinate and then multiplication by ab; iteration gives all polynomials in ab. The displayed spaces are invariant, proving both inclusions. Hence the two actual end groups are $N_0 \rtimes D$ and $N_3 \rtimes D$.

If $a \in K$, then $b \in K$ and $N_0 = N_3$, so the end intersection is larger than D and IP fails. Otherwise $aK \cap K = 0$ and $bK \cap K = 0$, giving $N_0 \cap N_3 = 0$ and end intersection exactly D. Each end has rank-three IP: for example $\langle r_0, r_1 \rangle$ has translation space $\mathrm{span}_{\mathbb{F}_2}\{(0, 1), (a, 0)\}$ and meets D in $\langle r_1 \rangle$. Its outer adjacent order is four, and the other end follows by reversal. The end criterion proves full IP. The sum $N_0 \oplus N_3$ has dimension four over K, which gives the group and type claimed.

Finally a nondegenerate marking of rank at least five has a rank-four initial segment of this form. Its fifth generator commutes with r_0 and r_2, which have a common axis and distinct centres. By Lemma 1, it must have that same axis. It also commutes with r_1, whose axis is different and whose centre is not on the common axis. This is impossible by the same lemma. □

Scope and attribution

Theorems 2, 3, and 4 recover the source's Theorems 5.4, 5.5, and 5.6, and the corresponding rank-three and rank-four rows of Table 2. In particular the notation for the elementary abelian groups refers to the exact trace fields displayed above; it does not identify an arbitrary divisor of the ambient field order with a subfield. The classification of rank-three projective-line C-markings remains its separately attributed earlier result. The present argument classifies their possible group, without repeating that census. The rank-two semisimple rows of Table 2 and the odd-characteristic classifications are outside this component. No chiral rank-six nonexistence result is asserted here: that argument still has to join the actual two regular end actions.

Four homologies in a projective plane

A recovery of the odd-characteristic rank-four classification in L027

Let F be a finite field of odd characteristic p. We classify all rank-four string C-markings in $\mathrm{PSL}_3(F)$, including the degenerate case, and obtain the directly regular alternatives needed in Section 6 of Leemans–Vandenschrick. The normal forms originate in their Section 5.3. The proof below uses the actual translation kernels and an acquired projective-line classification, without first classifying all rank-three subgroups of $\mathrm{PSL}_3(F)$.

We retain the usual string end-intersection criterion, quoted as Lemma 2.1 of the source, and the classical identification $\mathrm{SO}_3(F) \cong \mathrm{PGL}_2(F)$. We also acquire the previously proved Topic E classification: a rank-four string C-marking of $\mathrm{PSL}_2(K)$ or $\mathrm{PGL}_2(K)$ is exactly the $\mathrm{PSL}_2(11)$ marking of type $[3,5,3]$, the $\mathrm{PSL}_2(19)$ marking of type $[5,3,5]$, or the $\mathrm{PGL}_2(5)$ marking of type $[3,3,3]$. Its finite presentations and complete intersection checks remain in that acquired component. Dickson's subgroup theorem, in its actual natural-subfield form, is the only classification input used in passing to those groups.

The diagonal outer pair

Lemma 1. *A string C-marking in* $\mathrm{PSL}_3(F)$ *has rank at most four. Every rank-four marking has, after a projective change of coordinates, unique order-two lifts of the form*

$$r_0 = \mathrm{diag}(1,-1,-1), \qquad r_3 = \mathrm{diag}(-1,-1,1),$$
$$r_1 = \begin{pmatrix} a & b & 0 \\ c & -a & 0 \\ 0 & 0 & -1 \end{pmatrix}, \qquad r_2 = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -d & f \\ 0 & e & d \end{pmatrix}, \qquad bc = 1-a^2, \quad ef = 1-d^2.$$

Here $a, d \notin \{0,-1\}$. *If either is one, exactly one of its two off-diagonal parameters is zero.*

Proof. The scalar centre has odd order dividing three, so projective involutions lift uniquely to involutions of determinant one. Projectively commuting lifts commute: a scalar commutator ζ would have $\zeta^2 = 1$. In rank at least five, the first, third and fifth generators give an independent C_2^3. Their lifts are simultaneously diagonalizable, whereas the determinant-one diagonal sign group has order four. This contradicts independence.

At rank four, simultaneously diagonalize the independent outer pair as r_0, r_3 above. The involution r_1 preserves both eigenspaces of r_3. It acts as -1 on $\langle e_3 \rangle$, since otherwise its one-dimensional $+1$ eigenspace would be that point and $r_1 = r_3$. Its remaining block has trace zero and determinant -1. The same argument applies to r_2, giving the displayed forms. If $a = 0$, then $(r_0 r_1)^2 = r_3$, contrary to IP. If $a = -1$, then $(r_0 r_1)^p = r_3$, again impossible. If $a = 1$ and $b = c = 0$, then $r_1 = r_0$. Reversal gives the same assertions for d, e, f. □

The conic alternative

Lemma 2. *A dihedral group has no nondegenerate rank-three string C-marking.*

Proof. All three generators must be reflections: a central involution would commute with its neighbour. Write the commuting distinct ends as r and rz, where z is the central involution.

If s is the middle reflection, then $(srz)^2 = (rs)^{-2}$. Since $|rs| > 2$, this is a nonidentity rotation in both dihedral ends and not in $\langle s \rangle$, contradicting IP. □

Lemma 3. *If all four off-diagonal parameters b, c, e, f are nonzero, the marking is one of the three acquired projective-line markings. The ambient characteristics are respectively* $11, 19, 5$.

Proof. Direct multiplication shows that the four generators preserve the nonsingular diagonal quadratic form

$$Q = (c/b)x_1^2 + x_2^2 + (f/e)x_3^2.$$

The resulting orthogonal group has trivial intersection with the scalar centre of $\mathrm{SL}_3(F)$, since a scalar preserving Q has square one. Thus the marked group embeds faithfully in $\mathrm{PGL}_2(F)$. Its outer labels exceed two by Lemma 1, and the middle generators do not commute: the $(1,3)$ entry of $r_1 r_2$ is $bf \neq 0$, whereas that of $r_2 r_1$ is zero. The marking is nondegenerate.

We apply Dickson's subgroup alternatives to this actual subgroup. Cyclic groups and the involution-generated subgroup V_4 of A_4 are excluded. In an odd-characteristic affine group every involution has the form $x \mapsto -x + v$, and two distinct such involutions do not commute; the commuting first and third generators exclude that case. The dihedral case is excluded by Lemma 2 applied to an end. If the whole group were S_4 or A_5, either proper rank-three end would lie in an A_4, a dihedral subgroup or an abelian subgroup, by their elementary proper-subgroup lists; none admits a nondegenerate rank-three marking. IP makes these ends proper, so these two cases are excluded as well.

The group is therefore a natural subfield $\mathrm{PSL}_2(K)$ or $\mathrm{PGL}_2(K)$. The acquired Topic E theorem gives the three marked possibilities. The orders and simplicity distinguish them: $\mathrm{PSL}_2(11)$ and $\mathrm{PSL}_2(19)$ are simple, while $\mathrm{PGL}_2(5)$ has an index-two subgroup. Their natural subfield embeddings consequently have characteristic $11, 19, 5$, respectively. Conversely their standard three-dimensional orthogonal representations embed them in $\mathrm{PSL}_3(F)$ in those characteristics, with the same actual markings. □

The affine alternatives

Duality transposes each lift, interchanging b, c and also e, f. Reversal, followed by interchange of the first and third coordinates, interchanges the two pairs. These operations reduce all remaining cases to $b = 0$, with either no further zero, $e = 0$, or $f = 0$. The nonzero parameter c may be scaled to one; when $d = 1$, the other nonzero off-diagonal parameter can also be scaled to one. These are changes of basis, not changes to the ordered generated group.

Lemma 4. *The case $b = 0$ and $ef \neq 0$ cannot have IP.*

Proof. Here $a = 1$, $c \neq 0$, and $d \notin \{0, 1, -1\}$. In block form with the invariant plane $\langle e_2, e_3 \rangle$, write

$$r_1 = \begin{pmatrix} 1 & 0 \\ ce_1 & -I_2 \end{pmatrix}, \quad r_2 = \mathrm{diag}(-1, P), \quad r_3 = \mathrm{diag}(-1, Q_0).$$

Here

$$P = \begin{pmatrix} -d & f \\ e & d \end{pmatrix}, \qquad Q_0 = \mathrm{diag}(-1, 1).$$

Let $T(v) = \left(\begin{smallmatrix} 1 & 0 \\ v & I_2 \end{smallmatrix}\right)$. The actual subgroup $B = \langle r_1, r_2, r_3 \rangle$ contains $(r_1 r_2)^2 = T((1-d)c, ec)$. Its intersection with the additive translation group is invariant under $-P$ and $-Q_0$. The latter is the sign change of the second coordinate. Adding and subtracting its image and dividing by two projects onto each coordinate. Hence this intersection contains $T((1-d)c, 0)$. Applying $-P$ and projecting again multiplies that coordinate by d. Since $\mathbb{F}_p[d]$ is a field and $1 - d \neq 0$, its polynomial span contains $T(c, 0)$. But $r_1 = T(c, 0) r_0$, so $r_0 \in B$, contrary to IP. □

Theorem 5. *The two remaining affine cases are as follows.*

1. *If $b = e = 0$, the marking has type $[p, 2, p]$ and group $D_{2p} \times D_{2p}$.*
2. *If $b = f = 0$, the marking has type $[p, 2p, p]$ and group $\mathrm{Heis}(p) \rtimes C_2^2$, where the Heisenberg group consists of all lower unitriangular 3×3 matrices over $\mathbb{F}_p$.*

Each displayed normal form has full IP and is directly regular.

Proof. In the first case, after scaling $c = f = 1$, the two cyclic translation groups $\langle I + E_{21} \rangle$ and $\langle I + E_{23} \rangle$ commute. The diagonal r_0 inverts the first and fixes the second; r_3 does the reverse. This is the actual direct product of two dihedral groups, with its two rank-two markings placed consecutively. The type and all subset intersections follow from that direct product.

In the second case set $c = e = 1$. Put

$$h(x, y, z) = \begin{pmatrix} 1 & 0 & 0 \\ x & 1 & 0 \\ z & y & 1 \end{pmatrix}, \quad D_0 = r_0, \quad D_3 = r_3.$$

The root groups $h(x, 0, 0)$ and $h(0, y, 0)$, generated respectively by $r_0 r_1$ and $r_2 r_3$, have commutators giving all $h(0, 0, z)$, with $x, y, z \in \mathbb{F}_p$. The two diagonal generators normalize this Heisenberg group and intersect it trivially. The full group thus consists uniquely of $h(x, y, z) D_0^i D_3^j$, with $i, j \in \{0, 1\}$, and has order $4p^3$. Here

$$r_1 = h(1, 0, 0) D_0, \qquad r_2 = h(0, -1, 0) D_3.$$

The first actual end $A = \langle r_0, r_1, r_2 \rangle$ consists of these normal forms with $y = -j$, and the second $B = \langle r_1, r_2, r_3 \rangle$ consists of those with $x = i$. For example A contains the full x, z translation plane (conjugate the x root group by r_2), and its complement is the commuting pair r_0, r_2. This proves both inclusions in its displayed description; the other end follows in the same way.

Their intersection has $x = i, y = -j$ and arbitrary z, hence order $4p$. Direct multiplication gives $(r_1 r_2)^2 = h(0, 0, 1)$ and shows that $r_1 r_2$ has order $2p$. Consequently $A \cap B = \langle r_1, r_2 \rangle$. Within A, the subgroup $\langle r_0, r_1 \rangle$ has $j = y = z = 0$; intersecting it with the preceding description of $\langle r_1, r_2 \rangle$ leaves exactly $\langle r_1 \rangle$. Reversal gives the other rank-three test. The end criterion proves full IP. The outer products have order p, giving type $[p, 2p, p]$.

Finally, in both cases the map to C_2 which sends a diagonal $D_0^i D_3^j$ to $i + j$ sends all four generators to its nonidentity element. Its kernel is exactly the even-word subgroup, proving direct regularity. □

Classification and source qualifications

Combining the lemmas gives exactly the source's Theorem 5.3: the degenerate direct product, the Heisenberg extension, and the three conic markings. The nondegenerate directly regular possibilities are only the Heisenberg marking of type $[p, 2p, p]$ and the simplex marking of type $[3, 3, 3]$ in characteristic five. The two simple conic groups have no index-two subgroup and are not directly regular. The simplex marking is directly regular by the usual sign map on S_5.

The source's physical page 17 names D_{2p} in the two-zero direct product case; its Theorem 5.3 correctly gives D_{2p}^2. The actual matrix decomposition above settles this discrepancy. Its two-point description also requires allowing duality: our chosen form fixes two lines, and its dual fixes two points. The restriction "$q = 5$" in the rank-six discussion must mean characteristic five for this subgroup alternative, since the same natural subfield marking embeds for every extension of $\mathbb{F}_5$. This is not yet a proof of the rank-six exclusion. The source's Table 1 parameter calculations are replaced here by the acquired complete marking classification; no new finite census is charged to that previously recovered result. The broader rank-three classifications in Theorems 5.1–5.2 remain a separate source-contract item.

The two regular ends of a chiral marking

A recovery of Theorem 1.3 of Leemans–Vandenschrick

We prove that neither $\mathrm{PSL}_3(q)$ nor $\mathrm{PSU}_3(q)$, for $q \geq 2$, is the automorphism group of a chiral polytope of rank greater than five. The acquired classifications of regular subgroups give the two possible local actions. The argument must allow one end to preserve a point and the other a line. This distinction replaces an unjustified inference in the source's even-characteristic proof.

We use the standard group characterization of chiral polytopes, in the notation of Section 2 of the source, with $r \geq 3$. Thus $G = \langle \sigma_1, \ldots, \sigma_{r-1} \rangle$, every σ_i is nonidentity, and $\tau_{ij} = \sigma_i \cdots \sigma_j$ is an involution for $i < j$. For $I \subseteq \{0, \ldots, r-1\}$ put

$$G^I = \langle \tau_{ij} : i \leq j,\ i-1, j \in I \rangle.$$

The actual intersection property is $G^I \cap G^J = G^{I \cap J}$; in particular a singleton index set gives the trivial group. No quotient intersection property will be used.

The regular ends and a proper normal subgroup

Lemma 1. *The involutions $\tau_{12}, \ldots, \tau_{1,r-1}$ form a directly regular string C-marking of rank $r-2$, of type $[|\sigma_3|, \ldots, |\sigma_{r-1}|]$. There is a second such marking $\tau_{r-2,r-1}, \ldots, \tau_{1,r-1}$, with reversed type $[|\sigma_{r-3}|, \ldots, |\sigma_1|]$.*

Proof. For $J \subseteq \{2, \ldots, r-1\}$ the subgroup generated by τ_{1j} for $j \in J$ is exactly $G^{\{0\} \cup J}$. Indeed, each remaining generator with both endpoints in J is $\tau_{i,j} = \tau_{1,i-1}^{-1} \tau_{1,j}$. Actual IP therefore gives all subset intersections of the proposed marking. For $k - j \geq 2$, $\tau_{1j}\tau_{1k} = \tau_{j+1,k}$ is an involution, proving distant commutation. Adjacent products are σ_{j+1}, as asserted.

The even-word subgroup of any involutory marking is generated by its adjacent products. Here it is $\langle \sigma_3, \ldots, \sigma_{r-1} \rangle$, a normal subgroup of index at most two. It cannot contain τ_{12}: its intersection with $G^{\{0,2\}}$ is $G^{\{2\}} = 1$. Thus the index is two. Applying the same argument to the reversed inverse rotation tuple gives the second marking. This also proves the source's Lemma 2.3 directly from its actual intersection convention. □

Lemma 2. *If $r \geq 3$ and some σ_i has order two, then G has a proper nontrivial normal subgroup. Also σ_i and σ_j commute whenever $|i - j| > 2$.*

Proof. First note an elementary consequence of the string relations. If $B^2 = (AB)^2 = (BC)^2 = (ABC)^2 = 1$, then $BAB = A^{-1}$, $BCB = C^{-1}$, and

$$1 = (ABC)^2 = A(BCB)(BAB)C = AC^{-1}A^{-1}C.$$

Taking $A = \sigma_i$, $B = \sigma_{i+1} \cdots \sigma_{j-1}$ and $C = \sigma_j$ proves the distant commutation statement.

Now let $t = \sigma_i$ have order two, and put $L = \langle \sigma_1, \ldots, \sigma_{i-1} \rangle$ and $R = \langle \sigma_{i+1}, \ldots, \sigma_{r-1} \rangle$. Conjugation by t inverts the adjacent generator on either side. It centralizes every longer interval ending immediately before it or starting immediately after it, because both that interval and its product with t are involutions. These intervals and the adjacent generator generate the corresponding side, so t normalizes L, R. Distant commutation shows that the

two sides commute except possibly $\sigma_{i-1}, \sigma_{i+1}$. The identity above, with $B = t$, covers that pair as well. Moreover $L \cap R = 1$ by their disjoint vertex-index sets. Hence $H = LR$ is a normal direct product in G.

Here properness is an actual IP assertion, not an assumption of independence of rotation generators. If $t = lr$ with $l \in L, r \in R$, then

$$r \in \langle L, t \rangle \cap R = G^{\{0,\dots,i\}} \cap G^{\{i,\dots,r-1\}} = 1.$$

It follows that $t \in L \cap \langle t \rangle = G^{\{i-1\}} = 1$, a contradiction. The same calculation holds at an endpoint, with the empty side trivial. Since $r \geq 3$, at least one other nonidentity rotation belongs to H, so H is nontrivial. □

We acquire the complete even-characteristic regular subgroup theorem: a nondegenerate regular string marking in $\mathrm{PSL}_3(F)$ has rank at most four, and at rank four it has, up to coordinates and duality,

$$R_0 = I + E_{23}, \quad R_1 = I + aE_{31}, \quad R_2 = I + bE_{13}, \quad R_3 = I + E_{21}, \qquad ab \neq 0. \tag{1}$$

We also acquire the complete odd-characteristic theorem. Its regular rank bound is four without a nondegeneracy assumption. Its nondegenerate directly regular rank-four markings are the Heisenberg marking $[p, 2p, p]$ and the simplex marking $[3, 3, 3]$ in characteristic five. The other two conic markings have simple groups and are not directly regular. These are actual marked-group statements, not just group orders.

The classical simplicity and order formulas for the projective linear and unitary groups imply that $\mathrm{PSL}_3(q)$ is simple for $q \geq 2$, and $\mathrm{PSU}_3(q)$ is simple for $q \geq 3$. Also $\mathrm{PSU}_3(q)$ embeds in $\mathrm{PSL}_3(q^2)$: the intersection of the scalar centre of $\mathrm{SL}_3(q^2)$ with $\mathrm{SU}_3(q)$ is precisely the latter's centre. Lemmas 1 and 2, and the acquired rank bounds, consequently give $r \leq 6$ for these simple groups.

The small unitary group

Lemma 3. *In* $\mathrm{PSU}_3(2)$ *no two distinct involutions commute. In particular it cannot be the group of a chiral polytope of rank at least five.*

Proof. The scalar kernel has odd order three. Every projective involution therefore has a unique involutory lift, and projectively commuting involutory lifts commute: their scalar commutator has square one. In characteristic two a nonidentity involution of dimension three is $T = I + uf$, where $u \neq 0$, $f \neq 0$ and $f(u) = 0$. Let h be the nonsingular Hermitian form, linear in its second argument. For $x \in \ker f$, preservation of $h(x, y)$ by T gives $h(x, u)f(y) = 0$ for every y. Thus $\ker f = u^{\perp}$, and $f(x) = \lambda h(u, x)$ for a nonzero scalar λ. The equation $f(u) = 0$ makes u isotropic; preservation of h now reduces to $\lambda + \lambda^2 = 0$. Over $\mathbb{F}_4$ it has just one nonzero solution. Consequently each projective centre supports exactly one nonidentity unitary elation.

Two distinct commuting elations in a projective plane have the same centre or the same axis, by the acquired rank-one commutation lemma. Here the axis is the Hermitian polar of the centre, so either condition forces the same centre and hence the same involution. This proves the first claim. A rank-three string C-marking would contain distinct commuting first and third generators. Lemma 1 therefore rules out chiral rank at least five. We have not used, or re-proved, the source's stronger historical computation of no chiral markings at any rank. □

The mixed point and line case

Assume henceforth that a simple group above has a chiral marking of rank six. Define the actual regular ends

$$R = (\tau_{12}, \tau_{13}, \tau_{14}, \tau_{15}), \qquad S = (\tau_{45}, \tau_{35}, \tau_{25}, \tau_{15}). \tag{2}$$

Both are nondegenerate by Lemma 2. In even characteristic normalize R as in (1), using a single duality if necessary. The first, second and last marks of S are then

$$S_0 = R_1 R_3 = I + (e_2 + a e_3) e_1^t, \quad S_1 = R_0 R_3 = I + e_2 (e_1^t + e_3^t), \quad S_3 = R_3. \tag{3}$$

The marks S_0, S_3 have the same axis and different centres. The acquired rank-four normal form says that in its point-preserving orientation the outer pair has the same centre; in the dual orientation it has the same axis. Thus S is in the latter orientation. Its first and third marks share a centre, as do its second and fourth. Writing $u = e_2 + a e_3$, we obtain

$$S_2 = I + uf, \qquad f \neq 0, \qquad f(u) = 0. \tag{4}$$

These conclusions concern the actual axes and centres in (3); they do not infer a common fixed point of the two ends.

Since $\sigma_1 = R_3 S_2$ and $\sigma_5 = R_2 R_3$, the displayed lifts are

$$\Sigma_1 = I + E_{21} + uf, \qquad \Sigma_5 = I + b E_{13} + E_{21}. \tag{5}$$

Lemma 2 requires their projective commutation. The second lift is unipotent. If $\Sigma_1 \Sigma_5 = \zeta \Sigma_5 \Sigma_1$ for a scalar ζ, conjugacy of Σ_5 and $\zeta \Sigma_5$ forces $\zeta = 1$ by their eigenvalues. Thus the lifts themselves commute. But the first row of $\Sigma_5 \Sigma_1 - \Sigma_1 \Sigma_5$ is abf, which is nonzero. This excludes even characteristic.

The two odd-characteristic possibilities

The two ends in (2) have a common first type entry $|\sigma_3|$. If one is a simplex marking, the ambient characteristic is five, and its entry three differs from the Heisenberg entry five. Thus either both are simplex markings or both are Heisenberg markings.

In the simplex case all five σ_i have order three. The classical rotation presentation of the Coxeter group $[3,3,3,3,3]$ identifies its rotation group with A_7. Hence G is a nontrivial quotient of the simple group A_7 and has order 2520. This is impossible here: $|\operatorname{PSL}_3(2)| = 168$, $|\operatorname{PSL}_3(3)| = 5616$, and for $q \geq 4$

$$|\operatorname{PSL}_3(q)| = \frac{q^3(q^3-1)(q^2-1)}{(3, q-1)} \geq 20160.$$

Similarly $|\operatorname{PSU}_3(2)| = 72$, $|\operatorname{PSU}_3(3)| = 6048$, and for $q \geq 4$ its order is at least $4^3(4^3+1)(4^2-1)/3 = 20800$.

Finally suppose both ends are Heisenberg markings. Up to coordinates and a single duality, the acquired normal form gives

$$R_0 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \quad R_1 = \begin{pmatrix} 1 & 0 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix},$$

$$R_2 = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 1 & 1 \end{pmatrix}, \quad R_3 = \operatorname{diag}(-1, -1, 1).$$

These are the unique involutory determinant-one lifts. Therefore

$$S_0 = \begin{pmatrix} -1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \quad S_1 = \operatorname{diag}(-1, 1, -1), \quad S_3 = R_3.$$

The first two marks of S have common centre e_2 and distinct axes. In the displayed Heisenberg normal form the first pair has a common axis and the last pair a common centre; in its dual these roles reverse. Consequently the last pair of S has common axis $x_3 = 0$. This accounts for the dual orientation explicitly, and yields

$$S_2 = \begin{pmatrix} -1 & 0 & u \\ 0 & -1 & v \\ 0 & 0 & 1 \end{pmatrix}.$$

Commutation with S_0 gives $u = 2v$, although the remaining contradiction does not need this extra restriction. We have

$$\Sigma_1 = R_3 S_2 = I - uE_{13} - vE_{23}, \qquad \Sigma_5 = R_2 R_3 = I - E_{32}.$$

Again projective distant commutation lifts to ordinary commutation because Σ_5 is unipotent. Its matrix difference is

$$\Sigma_1 \Sigma_5 - \Sigma_5 \Sigma_1 = uE_{12} + vE_{22} - vE_{33}.$$

Thus $u = v = 0$, making σ_1 trivial, a contradiction. Together with Lemma 3 this completes the claimed rank bound.

Attribution and exact scope

The nonexistence result is Theorem 1.3 of the fixed 2022 source, and Lemmas 1 and 2 reprove its Lemmas 2.3,6.1,6.2. The actual regular subgroup classifications are separately acquired proofs of Section 5. The even argument treats both point and line orientations; the odd argument justifies the homology-axis inference and accounts for the projective scalar in distant commutation. The source's simplex phrase "$q = 5$" is read as characteristic five for this subgroup alternative. The rank bound includes $\mathrm{PSU}_3(2)$ by the elementary unitary argument above, without importing an unspecified computer search. The broader rank-three subgroup classifications, the source's small negative chiral censuses, and its remaining unitary existence questions are separate obligations, not consequences claimed here.

Small regular subgroups and their translation kernels

A recovery of Sections 5.1–5.2 of Leemans–Vandenschrick

Let $F = \mathbb{F}_q$ have characteristic p. We first recover the rank-two order bound in Section 5.1, with its unipotent correction. For odd p we then recover the reducible and conic rank-three alternatives of Theorems 5.1–5.2. The actual translation modules also give corrected types for the generic affine row of Table 3. We retain the standard string end-intersection criterion, and Dickson's classification of subgroups of $\mathrm{PGL}_2(F)$ in its natural subfield form. The latter is an attributed classical input, not a new classification claimed here.

Two involutions

Theorem 1. *If distinct involutions generate a rank-two subgroup of* $\mathrm{PSL}_3(q)$*, its order divides* $2(q-1)$*,* $2(q+1)$ *or* $4p$*. In characteristic two, the last possibility has rotation order at most four. In odd characteristic a regular unipotent rotation has order* p*.*

Proof. The scalar centre has odd order dividing three. Each projective involution has a unique involutory lift to determinant one, and the product X of the two lifts is conjugate to X^{-1}. Its eigenvalues are therefore $1, \lambda, \lambda^{-1}$. If $\lambda \neq \pm 1$, the matrix is semisimple: either $\lambda \in F$ and its order divides $q-1$, or $\lambda^q = \lambda^{-1}$ and its order divides $q+1$. If all eigenvalues are one, the largest Jordan block has size at most three. Its order divides p for odd p, and four for $p = 2$. In odd characteristic the remaining case has eigenvalues $1, -1, -1$; its order divides $2p$. Projectivization does not shorten any of these orders: if $X^m = \zeta I$, its eigenvalue one forces $\zeta = 1$. The group generated by the two involutions is dihedral of twice the rotation order, giving the assertion.

These are the expected ambient dihedral families, not just numerical bounds. The usual split and nonsplit tori in $\mathrm{PGL}_2(q)$ have normalizers $D_{2(q-1)}$ and $D_{2(q+1)}$; their three-dimensional orthogonal representation embeds in $\mathrm{PSL}_3(q)$ when q is odd. When $q > 2$ is even the block embedding of $\mathrm{SL}_2(q)$ gives the same normalizers. For $q = 2$ the nonsplit construction gives D_6; the trivial split torus gives no rank-two marking. In odd characteristic, $\mathrm{diag}(-1,-1,1)(I+E_{12})$ has order $2p$ and is inverted by $\mathrm{diag}(1,-1,-1)$, giving D_{4p}. In characteristic two, $I + E_{12}$ and $I + E_{23}$ give D_8. Their dihedral subgroups with rotation order at least two realize the corresponding smaller orders. □

The source's displayed regular unipotent on physical page 11 is incorrectly assigned the bound $2(2,p)$. Its order is p for odd p and four for $p = 2$. The final dihedral order bound remains valid. This is an isomorphism and order classification; no assertion that all embeddings of a given dihedral group are conjugate is needed.

A common block form in odd characteristic

Now suppose p is odd. Put

$$d_1 = \mathrm{diag}(1,-1,-1), \quad d_2 = \mathrm{diag}(-1,1,-1), \quad d_3 = d_1 d_2.$$

The commuting distinct end generators of a rank-three marking generate the diagonal Klein group $D = \{1, d_1, d_2, d_3\}$ after a coordinate change. The middle generator is a homology, with

a centre and a complementary axis. Suppose first that the whole group fixes a point or line. The common eigenspaces of D are exactly the coordinate points and coordinate planes. After duality if necessary, the middle axis therefore contains a coordinate point. Move this point to e_3. The middle lift has the form

$$t = \begin{pmatrix} A & 0 \\ v & -1 \end{pmatrix}, \qquad A = \begin{pmatrix} a & b \\ c & -a \end{pmatrix}, \quad a^2 + bc = 1, \quad vA = v. \tag{1}$$

The group is $G = \langle D, t \rangle$, regardless of which two nonidentity members of D are used as its ordered outer generators. Every such group fixes e_3 as a projective point. Its matrix lifts have bottom eigenvalue ± 1, so they meet the scalar centre trivially. We may work throughout with these actual matrix groups.

Write

$$M(x, y) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ x & y & 1 \end{pmatrix}.$$

A pure block matrix with upper block B and lower scalar $\delta = \det B \in \{1, -1\}$ acts by conjugation on these rows as $(x, y) \mapsto \delta(x, y)B$. This elementary action, together with coordinate sign projections, accounts for the generic cases.

Theorem 2. *Suppose $bc \neq 0$. Normalize*

$$A = \begin{pmatrix} a & -a - 1 \\ a - 1 & -a \end{pmatrix}, \qquad a \neq \pm 1. \tag{2}$$

Then v is either zero or can be normalized to $(1, -1)$. Put $K = \mathbb{F}_p(a)$, choose λ with $\lambda + \lambda^{-1} = 2a$, and set

$$h = |\lambda|, \quad h' = |-\lambda|, \quad m = \operatorname{lcm}(h, 2).$$

If $v = 0$, the generated group is D_{2m}. If $v = (1, -1)$, it is $K^2 \rtimes D_{2m}$ in the displayed faithful two-dimensional action. All three choices of distinct outer generators in D then give full rank-three IP. Up to reversal their types are

$$[h, h'], \qquad [h, 2p], \qquad [h', 2p]. \tag{3}$$

At $a = 0$ the group is $D_{2p} \wr C_2$ and these types reduce to $[4, 4]$ and $[4, 2p]$.

Proof. Diagonal scaling gives (2), and its left $+1$ eigenspace is $\langle (1, -1) \rangle$. Scaling the third coordinate gives the asserted normalization of v. Let $h_A = \operatorname{diag}(A, -1)$. For $v = (1, -1)$, the actual kernel of the upper-block action contains

$$(td_3)^2 = M(2, -2).$$

Conjugating by D and adding or subtracting the results separates the two coordinates. Dividing by two is valid in odd characteristic. Conjugation by t and projection then multiply a coordinate by a. Iteration gives every element of K in each coordinate. Conversely all matrices have entries in K. Thus this kernel is exactly $M(K^2)$. Since $t = M(v)h_A$, the group contains the pure block lift h_A and is the actual semidirect product $M(K^2) \rtimes H$, where $H = \langle D, h_A \rangle$. In the zero case it is just H.

The upper block of $d_1 h_A$ has determinant one and eigenvalues λ, λ^{-1}. They are distinct and not ± 1. Adjoining d_3, whose upper block is $-I$, enlarges its cyclic group to order m: for even h, its half-power is already $-I$, whereas for odd h adjoining $-I$ doubles the order. The remaining reflection inverts this cyclic group. Hence $H = D_{2m}$. In particular its action on K^2 is faithful and no nonidentity rotation fixes a nonzero vector. The trace field is K; equivalently it is the least field in which $\lambda + \lambda^{-1}$ lies. Thus m divides $|K| - 1$ or $|K| + 1$.

For completeness we prove the actual intersections. Put $A_i = \langle d_i, t\rangle$, and let $H_i = \langle d_i, h_A\rangle$ for $i = 1, 2$. With the convention $H^g = g^{-1}Hg$, direct multiplication gives

$$A_i = H_i^{M(c_i)}, \qquad c_1 = (0, (1-a)^{-1}), \quad c_2 = (-(a+1)^{-1}, 0). \tag{4}$$

The translations c_i fix d_i by conjugation and send h_A to t. The vectors differ. Therefore the common elements of these two complements inject into the stabilizer of $c_1 - c_2$ in H. Such a stabilizer has order at most two: a nonidentity rotation fixes no nonzero vector, and two reflections would have such a rotation as their product. Since both ends contain t, their intersection is exactly $\langle t\rangle$.

The third end is

$$A_3 = M(\mathbb{F}_p v) \rtimes \langle d_3, h_A\rangle, \qquad |A_3| = 4p. \tag{5}$$

Indeed its squared rotation gives $M(2v)$, which gives $M(v)$ and hence h_A. Its upper-block image is the Klein group $\langle d_3, h_A\rangle$. Each A_i, $i = 1, 2$, injects into that block action. If an element of $A_i \cap A_3$ had image d_3, equation (4) would force its translation to be $-2c_i$. This is not in the line Kv, and hence not in the kernel line of (5). The image $d_3 h_A$ is also excluded by multiplication by the common element t. Only the identity and h_A images remain, so $A_i \cap A_3 = \langle t\rangle$. The rank-three end criterion now proves full IP for every outer pair.

The rotations $d_1 t, d_2 t$ have orders h, h': their upper blocks have no eigenvalue one, so the affine displacement contributes nothing to their order. Equation (5) gives $|d_3 t| = 2p$. This proves (3). If $a = 0$, the block group is the signed permutation group on two coordinates. Its sign changes, together with $\mathbb{F}_p^2$, form two independent D_{2p} factors, exchanged by its coordinate swap. This is exactly the asserted wreath product. □

A dihedral group admits a rank-three string C-marking only in the degenerate case D_{4k} with k odd, of type $[k, 2]$ or its reverse. Here is the short argument needed for the zero case above. If all three marks are reflections, the commuting ends differ by the central involution; their common nontrivial squared rotation violates IP unless an adjacent product has order two. In that remaining situation a middle reflection commuting with an end is either that end or the other end, again violating independence. Thus the central involution is an end mark. The other two marks generate a dihedral subgroup intersecting it trivially, and the whole group is their direct product with C_2. This is dihedral exactly when their rotation order k is odd. The middle mark cannot be central, since then the whole group would be an elementary abelian group of rank at most two.

The triangular specializations

If $b = c = 0$ in (1), either $t \in D$, which fails IP, or the group is D_{4p}. If exactly one of b, c vanishes, interchanging the first two coordinates makes $a = 1$. Scaling the remaining entry and, when nonzero, the bottom row gives one of the following matrices, or the same D_{4p} case:

$$t_P = \begin{pmatrix} 1 & 0 & 0 \\ 1 & -1 & 0 \\ 1 & 0 & -1 \end{pmatrix}, \qquad t_H = \begin{pmatrix} 1 & -2 & 0 \\ 0 & -1 & 0 \\ 1 & -1 & -1 \end{pmatrix}. \tag{6}$$

For example when $b = 0, c \neq 0$, the equation $vA = v$ makes $v = (v_1, 0)$; when $c = 0, b \neq 0$, it makes $v_2 = bv_1/2$. This proves completeness of (6), including the cases with zero bottom row.

Theorem 3. *The group $\langle D, t_P\rangle$ is $D_{2p} \times D_{2p}$. Its two valid nondegenerate outer pairs have type $[p, 2p]$, up to reversal; the remaining pair fails IP. The group $\langle D, t_H\rangle$ is $\mathrm{Heis}(p) \rtimes C_2^2$. All three outer pairs have IP, with types $[p, 2p]$, $[p, 2p]$, $[2p, 2p]$, up to reversal.*

Proof. In the product case write a translation as $n(x,y) = I + xE_{21} + yE_{31}$. Then $t_P = n(1,1)d_1$. The diagonal signs separate its two prime-field coordinates: d_2 inverts the first, d_3 the second, and d_1 both. Thus the actual group is the direct product claimed. The end $A_1 = \langle d_1, t_P \rangle$ has normal forms $n(k,k)d_1^\epsilon$. In $A_2 = \langle d_2, t_P \rangle$ the coordinate x is arbitrary and y is zero or one according as its diagonal component fixes or negates the second coordinate. For A_3 the analogous condition holds on x. These descriptions follow by squaring $d_2 t_P, d_3 t_P$, and give $A_1 \cap A_2 = A_1 \cap A_3 = \langle t_P \rangle$, whereas $|A_2 \cap A_3| = 4$. The orders of $d_i t_P$ are respectively $p, 2p, 2p$.

For the second case use

$$U(x,y,z) = I + xE_{12} + yE_{31} + zE_{32}, \qquad U(x,y,z)U(X,Y,Z) = U(x+X, y+Y, z+Z+yX).$$

We have $t_H = U(2,1,1)d_1$. Conjugation by d_2 changes this unipotent factor to $U(-2,1,-1)$. Their commutator has central coordinate -4, so it generates the full prime-field central root group. Their product then supplies the second root group, and the first follows by division. This proves the actual full Heisenberg kernel and its diagonal complement D.

Put $A_i = \langle d_i, t_H \rangle$. The first end has forms $U(2k,k,k^2)d_1^\epsilon$. The kernels of the diagonal projections of the other two ends are respectively

$$N_2 = \{U(x,0,x/2) : x \in \mathbb{F}_p\}, \qquad N_3 = \{U(0,y,-y) : y \in \mathbb{F}_p\}.$$

These follow by squaring $d_2 t_H, d_3 t_H$; each end has order $4p$. The first kernel meets either of them trivially, giving $A_1 \cap A_2 = A_1 \cap A_3 = \langle t_H \rangle$. For the remaining intersection the kernels are disjoint. At diagonal component d_2, the possible translations in A_2, A_3 are respectively $U(x,0,x/2)$ and $U(2,t,t)$; equality would force $x = 2, t = 0$ and then $1 = 0$. At component d_3 they are $U(x-2,1,x/2-1)$ and $U(0,y,-y)$; equality would give $x = 2, y = 1$ and then $0 = -1$. Thus only the identity and d_1 components remain, with unique common lifts $1, t_H$. This proves the final intersection. The rotation orders are $p, 2p, 2p$, and the end criterion completes the proof. □

The irreducible conic alternative

Theorem 4. *If an odd-characteristic rank-three string C-group in* $\mathrm{PSL}_3(F)$ *fixes no projective point or line, it preserves a nonsingular conic. Its group is one of* $S_4, A_5, \mathrm{PSL}_2(K), \mathrm{PGL}_2(K)$, *in the natural three-dimensional orthogonal representation over a subfield* K.

Proof. With the diagonal outer Klein group, write the middle homology as $t = -I + 2uf$, where $f(u) = 1$. Every coordinate of u and f is nonzero: a zero coordinate of u would preserve the corresponding coordinate plane, and a zero coordinate of f the coordinate point. The diagonal quadratic form

$$Q = \mathrm{diag}(f_1/u_1, f_2/u_2, f_3/u_3)$$

is nonsingular, has $Q(u,u) = 1$, and is preserved by D and t. Its determinant-one orthogonal group meets the scalar centre trivially, and is isomorphic to $\mathrm{PGL}_2(F)$. Apply the retained Dickson theorem. Semisimple cyclic and dihedral subgroups preserve the torus axis in this orthogonal representation. The remaining cyclic or dihedral cases lie in a point stabilizer in the projective-line action, which preserves its corresponding isotropic point. The same holds for the Borel-type subgroups in Dickson's list. These are excluded by hypothesis. An A_4 cannot be generated by involutions, whose generated subgroup is only V_4. The stated alternatives remain. □

For S_4 the nondegenerate types are $[3,3]$ and $[3,4]$, up to reversal. Both adjacent orders cannot be four: the involution types in S_4 would give only a subgroup D_8. For A_5 the possible types are $[3,5]$ and $[5,5]$: a $[3,3]$ marking would be a quotient of the Coxeter group S_4. The

usual tetrahedral and projective cubical and icosahedral markings supply these small possibilities. The wider rank-three marking classifications within the projective-line groups remain separately attributed inputs; we do not repeat them here.

The source table and the corrected affine row

Theorems 2–4 recover the group alternatives of Theorems 5.1–5.2. They also give every non-degenerate affine type in Table 3, with one necessary correction. The generic row in that table prints only $[k, 4]$ and $[k, k]$. Its literal middle matrix (2) with bottom row $(1, -1, -1)$ instead gives the three types (3). For a concrete instance take $F = \mathbb{F}_5$ and $a = 2$. Then $h = 3, h' = 6$, $K = \mathbb{F}_5$, $m = 6$, and the group has order 300. The three actual C-markings have types $[3, 6]$, $[3, 10]$, $[6, 10]$. Their full subset intersections are checked in the supplied finite control; the uniform proof above gives them independently of that computation. This is a correction to a type row, not a refutation of the subgroup theorem or of the paper's principal chiral constructions and rank bound.

The centre–axis normalization and original matrices are due to Leemans–Vandenschrick. The proof here replaces the separate geometric case calculations by one block action, its translation kernel and actual complement intersections. No numerical compression or new conjecture resolution is asserted by this component.

The parameter restrictions in Table 1

A source-table calculation for Leemans–Vandenschrick

The first case of Section 5.3 of the source has odd characteristic and parameters $a, a' \notin \{0, 1, -1\}$. Its four involutions preserve a nonsingular conic. When both maximal rank-three subgroups are S_4 or A_5, Table 1 records the possible characteristics for the three adjacent orders. We recover every numerical cell below. This is a necessary parameter calculation; it neither proves the intersection property nor replaces the separately acquired classification of the resulting groups.

Write X for the characteristic-polynomial variable. Direct multiplication of the four displayed matrices on physical page 16 gives

$$(X-1)(X^2-2aX+1), \quad (X-1)(X^2+CX+1), \quad (X-1)(X^2-2a'X+1),$$
$$C = -aa' + a + a' + 1.$$

Thus the source's b in the last characteristic polynomial on physical page 17 must be a'. For end order three, $a = -1/2$; for end order five,

$$a = \frac{-1+t}{4} \quad \text{or} \quad \frac{-1-t}{4}, \qquad t^2 = 5.$$

The first assertion follows from the two primitive cube roots; the second from $2a = \zeta + \zeta^{-1}$ for a primitive fifth root. In the present nondegenerate parameter case characteristic five is excluded from an order-five end, since it would give $a = 1$.

The adjacent order in the middle is three, four or five precisely when, respectively,

$$C = 1, \qquad C = 0, \qquad C^2 - C - 1 = 0. \tag{1}$$

For the semisimple cases this follows from the reciprocal eigenvalue pair. In characteristic three or five the repeated eigenvalue one requires a qualification: the two central involutions are distinct, so their product is not the identity. Its nontrivial unipotent part in three dimensions has order three or five, respectively. Their scalar centres cannot shorten the order, because the matrices have eigenvalue one. Thus (1) remains valid in the relevant characteristics.

An order-four middle cannot be adjacent to an order-five end in S_4 or A_5: the former contains no element of order five and the latter none of order four. These are the dashes in the source table; they are not omitted equations. An order-four middle with two order-three ends would require $C = -1/4 = 0$, which is impossible.

For the other cells, substitute the three possible values of each end parameter. Symmetry in a, a' and replacement $t \mapsto -t$ reduce the nine source rows to the following four forms. The last two columns are possible characteristics; the actual equation in (1) must still hold for the chosen root t in the ambient field.

(a, a')	C	middle 3	middle 5
$(-1/2, -1/2)$	$-1/4$	5	11
$(-1/2, (-1 \pm t)/4)$	$(1 \pm 3t)/8$	$\varnothing$	59
$((-1 \pm t)/4, (-1 \pm t)/4)$	$(1 \pm 5t)/8$	19	$3, 11$
$((-1+t)/4, (-1-t)/4)$	$3/4$	$\varnothing$	19

For clarity, these restrictions follow by integer elimination rather than a search over fields. Reduce the left sides of (1) modulo $t^2 - 5$ and clear powers of two. Their integer numerators u_3, u_5, in the same four rows and using the plus choice of sign, are

	u_3	$\mathrm{Res}_t(t^2-5, u_3)$	u_5	$\mathrm{Res}_t(t^2-5, u_5)$
1	-5	25	-11	121
2	$3t-7$	4	$-9t-13$	-236
3	$5t-7$	-76	$27-15t$	-396
4	-1	1	-19	361

(2)

A common root in a field of odd characteristic forces that prime to divide the corresponding integer resultant. Factoring (2) gives exactly the displayed characteristic lists. Conversely, in each listed characteristic the two polynomials have a common root in a finite extension, and it does not force either end parameter into $\{0, 1, -1\}$. The supplied exact polynomial calculation checks those small gcd assertions and retains all nine signed rows. The ambient field must contain this common root; one may not assert that both choices of $\sqrt{5}$ work merely from the prime.

This recovers the parameter restrictions in the source table and corrects the last displayed characteristic polynomial. Its title must not be read as a certificate that every listed parameter choice gives a rank-four C-group. The actual generation and intersection conditions are supplied by the separately reviewed conic classification. The calculation adds no whole-original, conjecture, or measurement credit.

Three explicit unitary examples in rank four

A certificate for the examples in Section 7 of Leemans–Vandenschrick

The source reports three pairs of enantiomorphic chiral polytopes for $\mathrm{PSU}_3(17)$, of types $[3,7,3]$, $[3,8,3]$ and $[3,9,3]$. It gives neither matrices nor a search certificate for this sentence. We supply explicit matrices and verify these three positive examples. We do not infer that other types are absent, and we do not treat the source's historical question for larger fields as currently open.

Let $F = \mathbb{F}_{17}[\beta]$, where $\beta^2 = 3$. Since 3 is a nonsquare modulo 17, the nontrivial field automorphism is $\overline{\beta} = -\beta$. We use the standard Hermitian form $z^*z = \overline{z}^t z$. For a nonisotropic column z, put

$$R_z = -I + \frac{2zz^*}{z^*z}.$$

This matrix fixes z, negates its orthogonal complement, and has unitary determinant one. Put

$$u = R_{(4\beta,0,1)^t}, \qquad w = R_{(0,0,1)^t}.$$

For the three choices below, set $v = R_{(x,y,1)^t}$ and

$$a = wv, \qquad b = vwu, \qquad c = uw. \tag{1}$$

$\lvert b\rvert$	x	y	$\mathrm{Tr}_{F/\mathbb{F}_{17}}(x/(4\beta))$
7	$3+4\beta$	$1+3\beta$	2
8	$7+\beta$	$1+3\beta$	9
9	$8+6\beta$	$4+\beta$	3

(2)

Here and below orders are projective orders when the matrices represent unitary projective transformations.

Theorem 1. *For each row of (2), the images of (a,b,c) generate $\mathrm{PSU}_3(17)$ and satisfy the full string rotation intersection property. They define a rank-four chiral polytope of the indicated type $[3,|b|,3]$. The three types give three distinct pairs of enantiomorphic markings.*

The proof combines exact, small-input finite group calculations with a full-automorphism argument. We retain the standard rotation correspondence for abstract polytopes, the classical order and faithful isotropic-point action of $\mathrm{PSU}_3(17)$, and its classical automorphism description

$$\mathrm{Aut}(\mathrm{PSU}_3(17)) = \mathrm{PGU}_3(17) \rtimes \langle \overline{\ } \rangle. \tag{3}$$

For this prime field the last factor has order two. The inverse-transpose automorphism on unitary matrices is the same field map, so no separate graph coset is omitted. These are attributed classical group inputs, not new classifications asserted here. GAP's exact finite permutation group algorithms supply the group orders and subgroup intersections specified below; neither the source search nor a named-group constructor supplies them.

The interval relations and actual subgroup checks

All four nontrivial centres in (2) and the definition of u have Hermitian norm four. In particular all denominators are nonzero. The first and last rotations have order three. One may check this directly, or use the formula for two unitary involutions: their product has eigenvalue one and trace $4|z^*e_3|^2/(z^*z) - 1$. Here that trace is zero, so the remaining two eigenvalues are primitive cube roots. Equation (1) gives identically

$$ab = u, \qquad bc = v, \qquad abc = w. \tag{4}$$

The other required interval products are therefore involutions. Exact multiplication gives the middle orders in (2).

The finite calculation constructs all $4,914$ isotropic projective points directly. They are the normalized vectors

$$(1, s, t) \quad \text{with } 1 + s\bar{s} + t\bar{t} = 0, \qquad (0, 1, s) \quad \text{with } 1 + s\bar{s} = 0.$$

The remaining normalized point $(0, 0, 1)$ is not isotropic. These lists contain every projective point of norm zero exactly once. The program uses row representatives and right multiplication, preserving these Hermitian equations; the abstract matrix products are unchanged. The action of each literal matrix in (1) on these lists is computed, without identifying it with a preselected permutation generator. All three generated permutation groups have order

$$\frac{17^3(17^3 + 1)(17^2 - 1)}{3} = 2,317,678,272.$$

They are subgroups of the faithful projective unitary action; equality with its classical order proves generation of the whole group. The two adjacent subgroups have respective orders

$\lvert b\rvert$	$\lvert\langle a, b\rangle\rvert$	$\lvert\langle b, c\rangle\rvert$
7	168	168
8	2448	2448
9	2448	2448

No group identification from these last orders is needed.

We check every indexed intersection, rather than infer it from these orders. For $I \subseteq \{0, 1, 2, 3\}$ let G_I be generated by those of the following six elements whose endpoint pair lies in I:

$$(01) : a,\ (02) : ab,\ (03) : abc,\ (12) : b,\ (13) : bc,\ (23) : c.$$

The verifier constructs all sixteen actual permutation subgroups and checks $G_I \cap G_J = G_{I\cap J}$ for all 256 ordered pairs. For nested index sets it verifies actual subgroup containment; for all other pairs it computes the actual intersection. Adding the trivial endpoint conventions in the source does not change any of these groups. The three complete tests pass, giving precisely the rotation intersection property.

Excluding every reversing automorphism

Lemma 2. *Let u, w be the two matrices above. If $v = R_{(x,y,1)^t}$ and $\mathrm{Tr}_{F/\mathbb{F}_{17}}(x/(4\beta)) \neq 0$, no automorphism of $\mathrm{PSU}_3(17)$ fixes u, w and sends v to wvw.*

Proof. Write $\alpha = 4\beta$. A linear matrix commuting with w preserves its centre e_3 and its axis $\langle e_1, e_2\rangle$. Commuting also with u, whose centre has nonzero first and third coordinates, forces its two off-diagonal entries in the e_1, e_2 block to vanish and its first and third diagonal entries

to agree. Thus its projective form is $D_r = \mathrm{diag}(1, r, 1)$. It is a unitary similitude exactly when $r\bar{r} = 1$.

Fixing the projective involutions really does give these commuting matrix conditions: their determinant-one involutory lifts are unique, since the scalar centre has odd order three. Conjugation by a projective unitary representative, and the field map, preserve these lifts. The semilinear map given by field conjugation followed by $D_0 = \mathrm{diag}(\alpha/\bar{\alpha}, 1, 1)$ fixes u, w. Composing with it reduces every semilinear centralizer element to the linear case. Hence the full centralizer in (3) consists of the eighteen D_r and the eighteen maps $D_r D_0^{-}$.

The centre of wvw is $(-x, -y, 1)^t$. Its last coordinate is already normalized, so equality of centres under a linear centralizer element requires $x = -x$, hence $x = 0$. Under a semilinear element it requires $(\alpha/\bar{\alpha})\bar{x} = -x$, or

$$x/\alpha + \overline{x/\alpha} = 0.$$

Both possibilities contradict the hypothesis. This accounts for every automorphism in (3), independently of the second coordinate equation. □

A reversing automorphism for (1) would send a to a^{-1} while fixing $ab = u$ and $abc = w$, by the standard chiral criterion. Since $v = wa$, it would send v to $wa^{-1} = wvw$. The three traces in (2) are nonzero. The lemma therefore excludes all reversing automorphisms. With generation, (4) and the verified indexed intersections, the rotation correspondence proves the theorem. Distinct displayed types distinguish the three enantiomorphic pairs.

Certificate and scope

The supplied GAP program builds the field, the three homologies and all isotropic points afresh. It checks group membership, all exact rotation orders, all 768 indexed intersections and the 36 possible reversers for each row. An independent elementary field implementation checks the matrix identities, types and traces without importing GAP. Two additional controls prevent weaker tests from being mistaken for the theorem. The centre $(\beta, 1 + 2\beta, 1)$ gives the same type $[3, 8, 3]$ but only a group of order 2448, so fails generation. The centre $(1, 6, 1)$ has trace zero and admits an actual reversing semilinear map, so cannot pass the chirality test.

The exploratory normalization and sixteen completed intermediate cases are retained with their receipts. They are not an asserted census of all ranks or all types. In particular, our statement recovers the source's three reported positive pairs and does not turn the phrase "a computer search gave also three pairs" into an unproved assertion that there are no others. The positive claim is attributed to Leemans–Vandenschrick; the explicit centres and centralizer certificate are supplied here. No new question resolution, Lean proof, or numerical compression measurement is claimed.